

ELITE IGCSE ACADEMY

Edexcel IGCSE Mathematics A - Unit 2 (4WM2)

Higher Tier

Elite IGCSE Classified Unit 2 Expertise Answers (Q20+)

133 Questions | Question page then worked-solution page

PREPARED & CLASSIFIED BY

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Modular palette | Unit 2 / 4WM2

Student worked-solution book - Q20+ expertise.
Each question is followed by its worked solution on the next page.

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Direct & Inverse Proportion

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2022

Q#: 20

Marks: 4

TOPIC: **Direct & Inverse Proportion**

Source: Jan 2022 P2HR

- 20** y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 20

Direct & Inverse Proportion

1. Find inverse function

$$y \propto \frac{1}{\sqrt{x}}$$

2. Use Direct Inverse Proportion correct

and

$$x \propto T^3$$

3. Find inverse function

So

$$\sqrt{x} \propto T^{3/2}$$

4. Find inverse function

Therefore

$$y \propto \frac{1}{T^{3/2}}$$

5. Use Direct Inverse Proportion correct

So $yT^{3/2}$ is constant.

6. Find inverse function

When $y = 8$ and $T = 25$,

$$8(25)^{3/2} = 8(125) = 1000$$

7. Find inverse function

When $y = 27$,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

Final answer

$$\frac{100}{9}$$

Algebraic Fractions

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

Q#: 21

Marks: 3

TOPIC: **Algebraic Fractions**

Source: Specimen 4WM2H

$$21 \quad a = \frac{14}{3x-7} \quad x = \frac{7}{4y-3}$$

Express a in the form $\frac{py+q}{ry+s}$ where p , q , r and s are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM2H | Question 21

Algebraic Fractions

1. Simplify fraction

$$a = \frac{14}{3x - 7}, \quad x = \frac{7}{4y - 3}$$

2. Simplify fraction

Substitute for x :

$$a = \frac{14}{3\left(\frac{7}{4y-3}\right) - 7}$$

3. Simplify the denominator

Simplify the denominator:

$$\begin{aligned} 3\left(\frac{7}{4y-3}\right) - 7 &= \frac{21}{4y-3} - 7 \\ &= \frac{21 - 7(4y-3)}{4y-3} = \frac{42 - 28y}{4y-3} \end{aligned}$$

4. Simplify fraction

So

$$\begin{aligned} a &= \frac{14}{\frac{42-28y}{4y-3}} = \frac{14(4y-3)}{42-28y} \\ a &= \frac{4y-3}{3-2y} \end{aligned}$$

Final answer

$$a = \frac{4y-3}{3-2y}.$$

Algebraic Proof

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 20

Marks: 3

TOPIC: **Algebraic Proof**

Source: May 2025 4WM2H

20 | a, b and c are three consecutive even numbers where $a < b < c$

Prove algebraically that $b^2 = ac + 4$

(Total for Question 20 is 3 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 20
Algebraic Proof

1. Let

Let

$$a = 2n, \quad b = 2n + 2, \quad c = 2n + 4$$

2. Calculate value

Then

$$ac + 4 = (2n)(2n + 4) + 4$$

$$= 4n^2 + 8n + 4$$

$$= (2n + 2)^2 = b^2$$

3. Calculate valueSo $b^2 = ac + 4$.**Final answer**So $b^2 = ac + 4$.

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2020

Q#: 25

Marks: 3

TOPIC: **Algebraic Proof**

Source: Jan 2020 P2HR

25 N is a multiple of 5

$$A = N + 1$$

$$B = N - 1$$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

(Total for Question 25 is 3 marks)

EA

WORKED SOLUTION

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Jan 2020 P2HR | Question 25
Algebraic Proof

1. Use trigonometry

Since N is a multiple of 5, let

$$N = 5k$$

2. where is an integer

where k is an integer.

3. Given

Given

$$A = N + 1, \quad B = N - 1$$

4. Use trigonometry

So

$$A^2 - B^2 = (N + 1)^2 - (N - 1)^2$$

5. Expand

Expand:

$$(N + 1)^2 = N^2 + 2N + 1$$

$$(N - 1)^2 = N^2 - 2N + 1$$

6. Use trigonometry

Therefore

$$\begin{aligned} A^2 - B^2 &= (N^2 + 2N + 1) - (N^2 - 2N + 1) \\ &= 4N \end{aligned}$$

7. Substitute

Substitute $N = 5k$:

$$4N = 4(5k) = 20k$$

8. Use trigonometry

Since k is an integer, $20k$ is a multiple of 20.

Final answer

$A^2 - B^2$ is always a multiple of 20.

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2024

Q#: 23

Marks: 4

TOPIC: Algebraic Proof

Source: Nov 2024 P2H

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

WORKED SOLUTION

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Nov 2024 P2H | Question 23
Algebraic Proof

1. Evaluate fraction

$$\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x)$$

2. Factorise

Factorise:

$$16x^2 - 36 = 4(2x - 3)(2x + 3)$$

$$2x^2 + 7x + 6 = (2x + 3)(x + 2)$$

$$x^2 - 5x - 14 = (x - 7)(x + 2)$$

3. Evaluate fraction

So

$$\begin{aligned} & \frac{4(2x - 3)(2x + 3)}{x - 7} \div \frac{(2x + 3)(x + 2)}{(x - 7)(x + 2)} - (7 + 8x) \\ &= \frac{4(2x - 3)(2x + 3)}{x - 7} \times \frac{(x - 7)(x + 2)}{(2x + 3)(x + 2)} - (7 + 8x) \\ &= 4(2x - 3) - (7 + 8x) \\ &= 8x - 12 - 7 - 8x \\ &= -19 \end{aligned}$$

4. Calculate value

So $n = -19$, which is an integer.**Final answer**

$$n = -19$$

Solving Inequalities

EA

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

Source: May 2025 4WM2HR

20 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 20 is 3 marks)

EA

WORKED SOLUTION

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May 2025 4WM2HR | Question 20

Solving Inequalities

1. Required

Solve $2x^2 - 7x - 15 > 0$. Show clear algebraic working.

2. Factorise the quadratic 2 x

Factorise the quadratic:

$$2x^2 - 7x - 15 = (2x+3)(x-5)$$

$$\text{Check: } (2x+3)(x-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$$

3. Set each factor zero

Roots: set each factor to zero:

$$2x + 3 = 0 \implies x = -\frac{3}{2}; \quad x - 5 = 0 \implies x = 5$$

4. Solve inequality

Since the coefficient of x^2 is $+2 > 0$, the parabola opens **upward**. The expression is > 0 *outside* the roots:

Final answer

$$x < -\frac{3}{2} \text{ or } x > 5$$

Simultaneous Equations

EA

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: May 2025 4WM2HR

22 Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 22
Simultaneous Equations

1. Required

$$\text{Solve } y^2 + 5y + x^2 = 12 \text{ and } x = y + 5.$$

2. Substitute $x = y + 5$

Substitute $x = y + 5$ into the first equation :

$$y^2 + 5y + (y + 5)^2 = 12$$

3. Expand $(y + 5)^2$

$$\text{Expand } (y + 5)^2 = y^2 + 10y + 25 :$$

$$y^2 + 5y + y^2 + 10y + 25 = 12 \implies 2y^2 + 15y + 13 = 0$$

4. Factorise

Factorise:

$$(2y + 13)(y + 1) = 0 \implies y = -\frac{13}{2} \text{ or } y = -1$$

5. Find corresponding x values using

Find corresponding x values using $x = y + 5$:

Final answer

$$(x, y) = (4, -1) \text{ and } \left(-\frac{3}{2}, -\frac{13}{2}\right)$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 20

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Nov 2025 4WM2H

20 Solve the simultaneous equations

$$x^2 + y^2 = 3xy - 11$$

$$y = x + 2$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

EA

WORKED SOLUTION

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Nov 2025 4WM2H | **Question 20**
Simultaneous Equations

1. Substitute into equation**1. Substitute $y = (x+2)$ into the quadratic equation.**

$$x^2 + (x+2)^2 = 3x(x+2) - 11$$

2. Expand brackets

$$x^2 + (x^2 + 4x + 4) = 3x^2 + 6x - 11$$

$$2x^2 + 4x + 4 = 3x^2 + 6x - 11$$

3. Solve equation

$$0 = 3x^2 - 2x^2 + 6x - 4x - 11 - 4$$

$$x^2 + 2x - 15 = 0$$

4. Solve quadratic equation

$$(x+5)(x-3) = 0$$

So, $x = -5$ or $x = 3$.

5. Find corresponding y values using $y = x + 2$.

$$\text{If } x = -5 \implies y = -5 + 2 = -3$$

$$\text{If } x = 3 \implies y = 3 + 2 = 5$$

Final answer

$(-5, -3)$ and $(3, 5)$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

Q#: 20

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Specimen 4WM2H

20 Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

WORKED SOLUTION

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Specimen 4WM2H | Question 20
 Simultaneous Equations
1. Solve simultaneous equations

From

$$x - 2y = 3$$

$$x = 2y + 3$$

2. Solve simultaneous equationsSubstitute into $x^2 - y^2 + 2x = 10$:

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

3. Solve simultaneous equations

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

4. Solve simultaneous equationsIf $y = -\frac{1}{3}$,

$$x = 2\left(-\frac{1}{3}\right) + 3 = \frac{7}{3}$$

5. Solve simultaneous equationsIf $y = -5$,

$$x = 2(-5) + 3 = -7$$

Final answer
 $\left(\frac{7}{3}, -\frac{1}{3}\right)$ and $(-7, -5)$.

Solving Inequalities

EA

CLASSIFIED PROBLEM

Paper: P2H

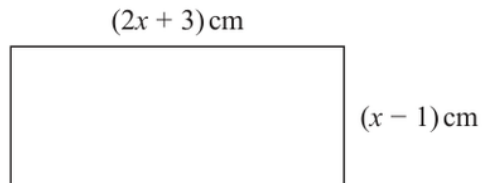
Session: Jan 2022

Q#: 22

Marks: 5

TOPIC: **Solving Inequalities**

Source: Jan 2022 P2H

22 Here is a rectangle.Diagram **NOT**
accurately drawnGiven that the area of the rectangle is less than 75 cm^2 find the range of possible values of x

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 22
Solving Inequalities

1. The rectangle has side lengths

The rectangle has side lengths

$$2x + 3$$

2. Use Functions Solving Inequalities

and

$$x - 1$$

3. Solve inequality

For the side lengths to be possible,

$$x > 1$$

4. The area is less than

The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

5. Factorise

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

6. Evaluate fraction

So

$$-\frac{13}{2} < x < 6$$

7. Solve inequality

Combining with $x > 1$:

$$1 < x < 6$$

Final answer

$$1 < x < 6$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 20

Marks: 4

TOPIC: **Solving Inequalities**

Source: Jun 2024 P2HR

- 20** Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | **Question 20**
Solving Inequalities

1. Solve quadratic equation

$$6x^2 - 7x - 20 > 0$$

2. Factorise

Factorise:

$$6x^2 - 7x - 20 = (3x + 4)(2x - 5)$$

3. The critical values are

The critical values are

$$x = -\frac{4}{3} \quad \text{and} \quad x = \frac{5}{2}$$

4. Solve quadratic equationSince the coefficient of x^2 is positive, the quadratic is greater than 0 outside the roots.**Final answer**

$$x < -\frac{4}{3} \quad \text{or} \quad x > \frac{5}{2}$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: **Solving Inequalities**

Source: Jun 2025 P2HR

21 Solve the inequality $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

(Total for Question 21 is 3 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | **Question 21**
Solving Inequalities

1. Solve inequality

$$2x^2 - 7x - 15 > 0$$

2. Factorise

Factorise:

$$2x^2 - 7x - 15 = (2x + 3)(x - 5)$$

3. The critical values are

The critical values are

$$x = -\frac{3}{2} \quad \text{and} \quad x = 5$$

4. The quadratic is greater than

The quadratic is greater than 0 outside the roots.

Final answer

$$x < -\frac{3}{2} \text{ or } x > 5$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2024

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

Source: Nov 2024 P2H

- 20** Solve the inequality $10x^2 + 11x - 21 < 0$
Show clear algebraic working.

(Total for Question 20 is 3 marks)

EA

WORKED SOLUTION

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Nov 2024 P2H | **Question 20**
Solving Inequalities

1. Solve inequality

$$10x^2 + 11x - 21 < 0$$

2. Factorise

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

3. The critical values are

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

4. The quadratic is less than

The quadratic is less than 0 between the roots.

Final answer

$$-\frac{21}{10} < x < 1$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2025

Q#: 25

Marks: 3

TOPIC: **Solving Inequalities**

Source: Nov 2025 P1H

- 25** Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

EA

WORKED SOLUTION

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Nov 2025 P1H | **Question 25**
Solving Inequalities

1. Solve inequality

$$2x^2 + x - 28 > 0$$

2. Factorise

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

3. The critical values are

The critical values are

$$x = \frac{7}{2} \quad \text{and} \quad x = -4$$

4. The quadratic is greater than

The quadratic is greater than 0 outside the roots.

Final answer

$$x < -4 \text{ or } x > \frac{7}{2}$$

Sequences

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 24

Marks: 4

TOPIC: **Sequences**

Source: May 2025 4WM2H

- 24** The 21st term of an arithmetic series is 109
The sum of the first 52 terms of the series is 4381
Work out the 5th term of the series.
Show clear algebraic working.

.....
(Total for Question 24 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 24
Sequences

1. Use series formula

For an arithmetic series,

$$a + 20d = 109$$

2. Evaluate fraction

and

$$\frac{52}{2}(2a + 51d) = 4381$$

$$26(2a + 51d) = 4381$$

3. Solving gives

Solving gives

$$d = -4.5, \quad a = 199$$

4. The fifth term is

The fifth term is

$$a + 4d = 199 + 4(-4.5) = 181$$

Final answer

181.

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

Q#: 24

Marks: 6

TOPIC: Sequences

Source: May 2025 4WM2HR

24 An arithmetic series P has first term a and common difference d

The sum of the first 40 terms of P is 2070

An arithmetic series Q has first term $2a$ and common difference $3d$

The 25th term of Q is 186

Work out the sum of the first 40 terms of Q

Show clear algebraic working.

(Total for Question 24 is 6 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 24
Sequences

1. Required

Series P : first term a , c.d. d ; $S_{40}(P) = 2070$. Series Q : first term $2a$, c.d. $3d$; 25th term of $Q = 186$. Find $S_{40}(Q)$.

2. Form equation from S_{40}

Form equation from $S_{40}(P) = 2070$:

$$S_{40} = \frac{40}{2}[2a + 39d] = 20(2a + 39d) = 2070 \implies 2a + 39d = 103.5 \quad (1)$$

3. Form equation from 25th term

Form equation from 25th term of $Q = 186$:

$$T_{25}(Q) = 2a + 24(3d) = 2a + 72d = 186$$

4. Solve simultaneous equations

Solve simultaneously – subtract (1) from (2):

$$33d = 82.5 \implies d = 2.5$$

5. Substitute $d = 2.5$ into (1)

Substitute $d = 2.5$ into (1) :

$$2a + 39(2.5) = 103.5 \implies 2a = 6 \implies a = 3$$

6. Compute $S_{40}(Q)$

Compute $S_{40}(Q)$, with first term $= 2a = 6$ and c.d. $= 3d = 7.5$:

$$S_{40}(Q) = \frac{40}{2}[2(6) + 39(7.5)] = 20[12 + 292.5] = 20 \times 304.5 = 6090$$

Final answer

$$S_{40}(Q) = 6090$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 26

Marks: 5

TOPIC: Sequences

Source: Nov 2025 4WM2H

26 The first term of an arithmetic series is 10

The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N

Show your working clearly.

$N =$

(Total for Question 26 is 5 marks)

EA

WORKED SOLUTION

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Nov 2025 4WM2H | Question 26

Sequences

1. Find common difference

1. Find the common difference d .The n^{th} term is given by $u_n = a + (n-1)d$.

$$u_{20} = 10 + 19d = 86$$

$$19d = 76$$

$$d = \frac{76}{19} = 4$$

2. Use sum formula

2. Use the sum formula to form an equation in terms of N .

$$S_N = \frac{N}{2} [2a + (N-1)d]$$

Substitute the known values $S_N = 5194$, $a = 10$, and $d = 4$:

$$5194 = \frac{N}{2} [2(10) + (N-1)(4)]$$

3. Solve quadratic equation

3. Simplify and solve the quadratic equation.

Multiply by 2 to clear the fraction:

$$10388 = N[20 + 4N - 4]$$

$$10388 = N(4N + 16)$$

$$4N^2 + 16N - 10388 = 0$$

4. Divide the entire equation by

Divide the entire equation by 4 to make it simpler:

$$N^2 + 4N - 2597 = 0$$

5. Use quadratic formula

4. Use the quadratic formula to find N .

$$N = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$N = \frac{-4 \pm \sqrt{16 - 4(1)(-2597)}}{2}$$

$$N = \frac{-4 \pm \sqrt{16 + 10388}}{2}$$

$$N = \frac{-4 \pm \sqrt{10404}}{2}$$

$$N = \frac{-4 \pm 102}{2}$$

6. Evaluate fraction

Since N must be a positive integer, we discard the negative root:

$$N = \frac{-4 + 102}{2} = \frac{98}{2} = 49$$

Final answer

$$N = 49$$

Always double-check that your N (number of terms) evaluates to a positive whole integer. If it ends up as a decimal or fraction, recheck your arithmetic progression formulas.

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

Q#: 24

Marks: 5

TOPIC: **Sequences**

Source: Specimen 4WM2H

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45
- Find the first term of this series.
Show clear algebraic working.

(Total for Question 24 is 5 marks)

WORKED SOLUTION

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Specimen 4WM2H | Question 24

Sequences

1. Let the first term beLet the first term be a and the common difference be d .

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

2. Find the nth term

So

$$S_{10} = 5(2a + 9d)$$

3. Find the nth term

and

$$S_5 = \frac{5}{2}(2a + 4d)$$

4. GivenGiven $S_{10} = 4S_5$:

$$5(2a + 9d) = 4\left(\frac{5}{2}(2a + 4d)\right)$$

$$10a + 45d = 20a + 40d$$

$$5d = 10a$$

$$d = 2a$$

5. The 8th term is

The 8th term is 45:

$$a + 7d = 45$$

6. SubstituteSubstitute $d = 2a$:

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

Final answer
the first term is 3.

EA

Simultaneous Equations

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 21

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2022 P2HR

21 Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

WORKED SOLUTION

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Jun 2022 P2HR | Question 21
Simultaneous Equations

1. Calculate value

$$x - 2y = 3$$

2. Calculate value

So

$$x = 2y + 3$$

3. Calculate value

Substitute into

$$x^2 - y^2 + 2x = 10$$

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

4. Evaluate fraction

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

5. FindFind x :

$$y = -\frac{1}{3} \Rightarrow x = \frac{7}{3}$$

$$y = -5 \Rightarrow x = -7$$

Final answer

$$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right) \text{ or } (-7, -5)$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2023

Q#: 21

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2023 P2H

21 Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 21

Simultaneous Equations

1. Calculate value

$$2x^2 + 3y^2 = 11$$

$$x = 3y - 1$$

2. Substitute

Substitute:

$$2(3y - 1)^2 + 3y^2 = 11$$

$$18y^2 - 12y + 2 + 3y^2 = 11$$

$$21y^2 - 12y - 9 = 0$$

$$7y^2 - 4y - 3 = 0$$

$$(7y + 3)(y - 1) = 0$$

3. Evaluate fraction

So

$$y = -\frac{3}{7} \quad \text{or} \quad y = 1$$

4. Find

Find x :

$$y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}$$

$$y = 1 \Rightarrow x = 2$$

Final answer

$$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right) \text{ or } (2, 1)$$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2024 P2HR

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 22

Simultaneous Equations

1. Calculate value

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

2. Substitute

Substitute:

$$x^2 + (3x - 1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x + 3)(2x - 3) = 0$$

3. Evaluate fraction

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

4. Find

Find y :

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

Final answer

$$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right) \text{ or } \left(\frac{3}{2}, \frac{7}{2}\right)$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2025 P1H

22 Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$

$$y = x + 2$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 22

Simultaneous Equations

1. Calculate value

$$x^2 + y^2 + 3y = 7$$

$$y = x + 2$$

2. Substitute

Substitute:

$$x^2 + (x + 2)^2 + 3(x + 2) = 7$$

$$x^2 + x^2 + 4x + 4 + 3x + 6 = 7$$

$$2x^2 + 7x + 3 = 0$$

$$(2x + 1)(x + 3) = 0$$

3. Evaluate fraction

So

$$x = -\frac{1}{2} \quad \text{or} \quad x = -3$$

4. Find

Find y :

$$x = -\frac{1}{2} \Rightarrow y = \frac{3}{2}$$

$$x = -3 \Rightarrow y = -1$$

Final answer

$$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right) \text{ or } (-3, -1)$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Jun 2025 P1HR

22 Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 22
Simultaneous Equations

1. Calculate value

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

2. Calculate value

So

$$y = 3 - 2x$$

3. Substitute

Substitute:

$$x^2 + (3 - 2x)^2 = 41$$

$$x^2 + 9 - 12x + 4x^2 = 41$$

$$5x^2 - 12x - 32 = 0$$

$$(5x + 8)(x - 4) = 0$$

4. Evaluate fraction

So

$$x = -\frac{8}{5} \quad \text{or} \quad x = 4$$

5. FindFind y :

$$x = -\frac{8}{5} \Rightarrow y = \frac{31}{5}$$

$$x = 4 \Rightarrow y = -5$$

Final answer

$$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right) \text{ or } (4, -5)$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2024

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Source: Nov 2024 P2H

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P2H | Question 22
Simultaneous Equations

1. Calculate value

$$x^2 + y^2 + y = 3$$

$$y = x + 2$$

2. Substitute

Substitute:

$$x^2 + (x + 2)^2 + (x + 2) = 3$$

$$x^2 + x^2 + 4x + 4 + x + 2 = 3$$

$$2x^2 + 5x + 3 = 0$$

$$(2x + 3)(x + 1) = 0$$

3. Evaluate fraction

So

$$x = -\frac{3}{2} \quad \text{or} \quad x = -1$$

4. Find

Find y :

$$x = -\frac{3}{2} \Rightarrow y = \frac{1}{2}$$

$$x = -1 \Rightarrow y = 1$$

Final answer

$$(x, y) = \left(-\frac{3}{2}, \frac{1}{2}\right) \text{ or } (-1, 1)$$

Forming & Solving Equations

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2025

Q#: 20

Marks: 3

TOPIC: **Forming & Solving Equations**

Source: Nov 2025 P2H

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c
Show your working clearly.

 $a = \dots\dots\dots$ $c = \dots\dots\dots$

(Total for Question 20 is 3 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 20

Forming & Solving Equations

1. Calculate value

For

$$ax^2 + 4x + c = 0$$

2. the quadratic formula gives

the quadratic formula gives

$$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$$

3. The given solutions are

The given solutions are

$$x = \frac{-4 \pm 2\sqrt{39}}{10}$$

4. Compare denominators

Compare denominators:

$$2a = 10$$

$$a = 5$$

5. Compare the square root parts

Compare the square root parts:

$$\sqrt{16 - 4ac} = 2\sqrt{39}$$

6. Square both sides

Square both sides:

$$16 - 4ac = 156$$

7. UseUse $a = 5$:

$$16 - 20c = 156$$

$$-20c = 140$$

$$c = -7$$

Final answer

$$a = 5, c = -7$$

EA

Functions

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 21

Marks: 7

TOPIC: **Functions**

Source: May 2025 4WM2H

21 The function g is defined as

$$g(x) = \frac{7x + 20}{2x} \quad x \neq 0$$

- (a) Solve $gg(x) = 6$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

P81641A

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The function h is defined as

$$h(x) = 5x^2 + 30x - 7 \quad x \geq -3$$

- (b) Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots (4)$$

(Total for Question 21 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 21
Functions

1. Evaluate fraction

(a) If $g(g(x)) = 6$, then

$$\frac{7g(x) + 20}{2g(x)} = 6$$

$$7g(x) + 20 = 12g(x)$$

$$g(x) = 4$$

2. Evaluate fraction

So

$$\frac{7x + 20}{2x} = 4$$

$$7x + 20 = 8x$$

$$x = 20$$

3. (b)

(b)

$$h(x) = 5x^2 + 30x - 7 = 5(x + 3)^2 - 52$$

4. Simplify surd

Let $y = 5(x + 3)^2 - 52$. Since $x \geq -3$, use the positive square root:

$$x + 3 = \sqrt{\frac{y + 52}{5}}$$

$$h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$$

Final answer

$$x = 20; h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3.$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 21

Marks: 7

TOPIC: Functions

Source: Nov 2025 4WM2H

21 The functions f and g are defined as

$$f(x) = 5x - 3$$

$$g(x) = 4x^2 + 16x - 9 \quad \text{where } x \geq -2$$

(a) Find $g(-1)$

(1)

(b) Find $fg(x)$

Give your answer in its simplest form.

$$fg(x) = \dots\dots\dots$$

(2)

$$g(x) = 4x^2 + 16x - 9 \quad \text{where } x \geq -2$$

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$$g^{-1}(x) = \dots\dots\dots$$

(4)

(Total for Question 21 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 21

Functions

1. (a) Find $g(-1)$ (a) Find $g(-1)$:

$$g(-1) = 4(-1)^2 + 16(-1) - 9 = 4 - 16 - 9 = -21$$

2. Answer

$$a - 21$$

(b) Find $fg(x)$:

$$\begin{aligned} fg(x) &= f(g(x)) = 5(4x^2 + 16x - 9) - 3 \\ &= 20x^2 + 80x - 45 - 3 = 20x^2 + 80x - 48 \end{aligned}$$

3. Answer

$$b \ 20x^2 + 80x - 48$$

Let $y = 4x^2 + 16x - 9$. We must rearrange to make x the subject.

1. Complete the square for the x terms.

$$y = 4(x^2 + 4x) - 9$$

$$y = 4((x+2)^2 - 4) - 9$$

$$y = 4(x+2)^2 - 16 - 9$$

$$y = 4(x+2)^2 - 25$$

2. Rearrange for x .

$$y + 25 = 4(x + 2)^2$$

$$\frac{y + 25}{4} = (x + 2)^2$$

$$x + 2 = \pm \sqrt{\frac{y + 25}{4}}$$

Since $x \geq -2$, $x + 2$ is positive, so we take the positive root.

$$x + 2 = \frac{\sqrt{y + 25}}{2}$$

$$x = \frac{\sqrt{y + 25}}{2} - 2$$

3. Swap y for x to write the inverse function.**Final answer**

$$g^{-1}(x) = \frac{\sqrt{x+25}}{2} - 2$$

When finding the inverse of a quadratic, always complete the square. The restricted domain ($x \geq -2$) tells you whether to take the positive or negative square root!

Sequences

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2020

Q#: 25

Marks: 3

TOPIC: Sequences

Source: Jan 2020 P2H

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of k .

$k =$

(Total for Question 25 is 3 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 25

Sequences

1. Mario saves from 2021 up

Mario saves from 2021 up to and including 2070.

2. Number of years

Number of years:

$$2070 - 2021 + 1 = 50$$

3. This is an arithmetic sequence

This is an arithmetic sequence.

4. Use trigonometry

First term:

$$50$$

5. Common difference

Common difference:

$$k$$

6. Last term

Last term:

$$50 + 49k$$

7. Use the arithmetic series formula

Use the arithmetic series formula:

$$S_n = \frac{n}{2}(a + l)$$

$$33125 = \frac{50}{2} (50 + (50 + 49k))$$

$$33125 = 25(100 + 49k)$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = 25$$

Final answer

$$k = 25$$

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2020

Q#: 22

Marks: 4

TOPIC: Sequences

Source: Jan 2020 P2HR

- 22** The first term of an arithmetic series S is -6
The sum of the first 30 terms of S is 2865

Find the 9th term of S .

(Total for Question 22 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 22
Sequences

1. Find the n th term

First term:

$$a = -6$$

$$S_{30} = 2865$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2865 = \frac{30}{2}(2(-6) + 29d)$$

$$2865 = 15(-12 + 29d)$$

$$191 = -12 + 29d$$

$$29d = 203$$

$$d = 7$$

3. The 9th term is

The 9th term is

$$u_9 = a + 8d$$

$$u_9 = -6 + 8(7) = 50$$

Final answer

50

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2021

Q#: 24

Marks: 4

TOPIC: Sequences

Source: Jan 2021 P1H

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .
Show your working clearly.

 $a = \dots\dots\dots$

(Total for Question 24 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 24
Sequences

1. Evaluate fraction

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{2n} = n(2a + (2n-1)d)$$

2. Given

Given

$$S_{2n} = 4S_n$$

$$n(2a + (2n-1)d) = 4 \cdot \frac{n}{2}(2a + (n-1)d)$$

3. Divide by

Divide by n :

$$2a + (2n-1)d = 2(2a + (n-1)d)$$

$$2a + 2nd - d = 4a + 2nd - 2d$$

$$d = 2a$$

$$a = \frac{d}{2}$$

Final answer

$$a = \frac{d}{2}$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2021

Q#: 21

Marks: 6

TOPIC: Sequences

Source: Jan 2021 P1HR

- 21** The n th term of an arithmetic series is u_n where $u_n > 0$ for all n
The sum to n terms of the series is S_n

Given that $u_4 = 6$ and that $S_{11} = (u_6)^2 + 18$

find the value of u_{20}

(Total for Question 21 is 6 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 21
Sequences

1. Let

Let $u_6 = v$.

2. Use series formula

For an arithmetic series,

$$S_{11} = \frac{11}{2}(u_1 + u_{11})$$

3. The average of and is

The average of u_1 and u_{11} is u_6 , so

$$S_{11} = 11u_6 = 11v$$

4. Given

Given

$$S_{11} = (u_6)^2 + 18$$

$$11v = v^2 + 18$$

$$v^2 - 11v + 18 = 0$$

$$(v - 2)(v - 9) = 0$$

5. Calculate value

So

$$u_6 = 2 \quad \text{or} \quad u_6 = 9$$

6. Find the nth term

But $u_4 = 6$. If $u_6 = 2$, then the common difference is negative and later terms would not all be positive. Therefore,

$$u_6 = 9$$

$$d = \frac{u_6 - u_4}{2} = \frac{9 - 6}{2} = \frac{3}{2}$$

$$u_{20} = u_6 + 14d$$

$$u_{20} = 9 + 14\left(\frac{3}{2}\right) = 30$$

Final answer

$$u_{20} = 30$$

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2022

Q#: 26

Marks: 5

TOPIC: Sequences

Source: Jan 2022 P2H

- 26** An arithmetic series has first term a and common difference d , where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m
Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 26
Sequences

1. Calculate value

$$S_m = 39$$

$$S_{2m} = 320$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

3. Calculate value

Then

$$2S_m = m(2a + (m-1)d) = 78$$

4. Calculate value

and

$$S_{2m} = m(2a + (2m-1)d) = 320$$

5. Subtract

Subtract:

$$S_{2m} - 2S_m = m[(2m-1)d - (m-1)d]$$

$$320 - 78 = m(md)$$

$$242 = m^2d$$

$$242 = 2 \times 11^2$$

6. Square factor

Since d is prime and m^2 is a square factor,

$$d = 2, \quad m^2 = 121$$

$$m = 11$$

Final answer

$$d = 2, \quad m = 11$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2022

Q#: 25

Marks: 5

TOPIC: **Sequences**

Source: Jun 2022 P1HR

- 25** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 25
Sequences

1. Calculate value

$$S_{10} = 4S_5$$

2. Use

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$S_5 = \frac{5}{2}(2a + 4d) = 5(a + 2d)$$

3. Calculate value

So

$$5(2a + 9d) = 4 \cdot 5(a + 2d)$$

$$2a + 9d = 4a + 8d$$

$$d = 2a$$

4. The 8th term is 45

The 8th term is 45:

$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

Final answer

The first term is 3

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2023

Q#: 23

Marks: 5

TOPIC: Sequences

Source: Jun 2023 P2HR

23 Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first n terms of the sequence is -1914

Work out the value of n

Show your working clearly.

$n =$

(Total for Question 23 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 23

Sequences

1. The first three terms are

The first three terms are

$$8p, \quad 7p - 3, \quad 4p + 2$$

2. Find the n th term

For an arithmetic sequence, consecutive differences are equal:

$$(7p - 3) - 8p = (4p + 2) - (7p - 3)$$

$$-p - 3 = -3p + 5$$

$$2p = 8$$

$$p = 4$$

3. Find the n th term

So the first term is

$$8p = 32$$

4. Find common difference

and the common difference is

$$(7p - 3) - 8p = 25 - 32 = -7$$

5. The sum of the firstThe sum of the first n terms is

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$-1914 = \frac{n}{2}(64 - 7(n - 1))$$

$$-1914 = \frac{n}{2}(71 - 7n)$$

$$-3828 = n(71 - 7n)$$

$$7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 \pm 335}{14}$$

6. Only the positive value is

Only the positive value is possible:

$$n = 29$$

Final answer

$$n = 29$$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2025

Q#: 24

Marks: 5

TOPIC: Sequences

Source: Jun 2025 P1H

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a =$

$d =$

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 24
Sequences

1. Evaluate fraction

$$u_{20} = a + 19d = 123$$

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$

$$2a + 29d = 192$$

2. Double the first equation

Double the first equation:

$$2a + 38d = 246$$

3. Subtract

Subtract:

$$9d = 54$$

$$d = 6$$

$$a + 19(6) = 123$$

$$a = 9$$

Final answer

$$a = 9, d = 6$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2025

Q#: 24

Marks: 6

TOPIC: Sequences

Source: Jun 2025 P1HR

24 The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where x and y are constants.

The sum of the first 9 terms of the series is 216

Find the value of x and the value of y

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 24
Sequences

1. The first 3 terms are

The first 3 terms are

$$2x + 5, \quad 3y - 4, \quad 4x - 2$$

2. Equal differences give

Equal differences give

$$(3y - 4) - (2x + 5) = (4x - 2) - (3y - 4)$$

$$6y - 6x = 11$$

3. Use series formula

For an arithmetic series,

$$S_9 = 9(a + 4d)$$

4. Here

Here

$$a = 2x + 5$$

$$d = (3y - 4) - (2x + 5) = 3y - 2x - 9$$

5. Given

Given $S_9 = 216$:

$$9[(2x + 5) + 4(3y - 2x - 9)] = 216$$

$$12y - 6x = 55$$

6. Solve

Now solve

$$6y - 6x = 11$$

$$12y - 6x = 55$$

7. Subtracting gives

Subtracting gives

$$6y = 44$$

$$y = \frac{22}{3}$$

8. Evaluate fraction

Then

$$6\left(\frac{22}{3}\right) - 6x = 11$$

$$x = \frac{11}{2}$$

Final answer

$$x = \frac{11}{2}, y = \frac{22}{3}$$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2021

Q#: 23

Marks: 5

TOPIC: Sequences

Source: May 2021 P1H

- 23** The sum of the first N terms of an arithmetic series, S , is 292
The 2nd term of S is 8.5
The 5th term of S is 13

Find the value of N .

Show clear algebraic working.

 $N = \dots\dots\dots$

(Total for Question 23 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 23
Sequences

1. The 2nd term is

The 2nd term is

$$a + d = 8.5$$

2. The 5th term is

The 5th term is

$$a + 4d = 13$$

3. Subtract

Subtract:

$$3d = 4.5$$

$$d = 1.5$$

$$a = 8.5 - 1.5 = 7$$

4. Use

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$292 = \frac{N}{2}(14 + 1.5(N - 1))$$

$$292 = \frac{N}{2}(1.5N + 12.5)$$

$$3N^2 + 25N - 1168 = 0$$

$$N = \frac{-25 + 121}{6} = 16$$

Final answer

$$N = 16$$

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2023

Q#: 20

Marks: 6

TOPIC: Sequences

Source: May 2023 P1H

20 The sum of the first 80 terms of an arithmetic series, S , is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 20 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 20

Sequences

1. Evaluate fraction

$$S_{80} = 470$$

$$40(2a + 79d) = 470$$

$$2a + 79d = \frac{47}{4}$$

2. The 75th term is

The 75th term is

$$a + 74d = 14.5 = \frac{29}{2}$$

3. Double it

Double it:

$$2a + 148d = 29$$

4. Subtract

Subtract:

$$69d = 29 - \frac{47}{4} = \frac{69}{4}$$

$$d = \frac{1}{4}$$

$$a + 74\left(\frac{1}{4}\right) = \frac{29}{2}$$

$$a = -4$$

5. Evaluate fraction

Now

$$S_X = 171$$

$$171 = \frac{X}{2} \left(2(-4) + (X-1)\frac{1}{4} \right)$$

$$171 = \frac{X(X-33)}{8}$$

$$X^2 - 33X - 1368 = 0$$

$$X = \frac{33 + 81}{2} = 57$$

Final answer

$$X = 57$$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2024

Q#: 24

Marks: 6

TOPIC: Sequences

Source: May 2024 P1H

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84° The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 24
Sequences

1. The interior angles form an

The interior angles form an arithmetic sequence with first term 84 and common difference 4.

2. Use trigonometry

So the sum of the angles is

$$\begin{aligned} & \frac{n}{2}(2(84) + (n-1)4) \\ &= \frac{n}{2}(168 + 4n - 4) \\ &= 2n^2 + 82n \end{aligned}$$

3. The sum of the interior

The sum of the interior angles of an n -sided polygon is

$$180(n-2)$$

4. Calculate value

So

$$2n^2 + 82n = 180(n-2)$$

$$2n^2 - 98n + 360 = 0$$

$$n^2 - 49n + 180 = 0$$

$$(n-45)(n-4) = 0$$

5. Solve inequality

Since $n > 5$,

$$n = 45$$

6. Use trigonometry

Therefore the sum of the interior angles is

$$180(45-2) = 7740$$

Final answer
7740 degrees

CLASSIFIED PROBLEM

Paper: P1HR

Session: May 2024

Q#: 23

Marks: 4

TOPIC: Sequences

Source: May 2024 P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) , (x+2) , (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1HR | Question 23
Sequences

1. The first three terms are

The first three terms are

$$4x - 14, \quad x + 2, \quad 7x - 9$$

2. Equal differences give

Equal differences give

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$$

$$-3x + 16 = 6x - 11$$

$$9x = 27$$

$$x = 3$$

3. Find the nth term

So the sequence is

$$-2, 5, 12, \dots$$

4. with common difference

with common difference 7.

$$S_{40} = \frac{40}{2}(2(-2) + 39(7))$$

$$S_{40} = 20(-4 + 273) = 5380$$

Final answer

5380

CLASSIFIED PROBLEM

Paper: P1H

Session: MayNov 2020

Q#: 24

Marks: 4

TOPIC: Sequences

Source: May-Nov 2020 P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 24
Sequences

1. The arithmetic sequence is

The arithmetic sequence is

$$8, 15, 22, 29, 36, \dots$$

2. Calculate value

So

$$a = 8, \quad d = 7$$

3. The n th term is

The n th term is

$$u_n = 8 + 7(n - 1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

4. There are

There are

$$100 - 50 + 1 = 51$$

5. Find the n th term

terms from the 50th to the 100th inclusive.

$$\text{sum} = \frac{51}{2}(351 + 701)$$

$$= 26826$$

Final answer

26826

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2021

Q#: 20

Marks: 5

TOPIC: Sequences

Source: Nov 2021 P2H

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is $(90 + 2k)$,

calculate the sum of the first 30 terms of the series.

(Total for Question 20 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P2H | Question 20
Sequences

1. The first four terms are

The first four terms are

$$k, \quad \frac{3k}{4}, \quad \frac{k}{2}, \quad \frac{k}{4}$$

2. The common difference is

The common difference is

$$d = -\frac{k}{4}$$

3. The 15th term is

The 15th term is

$$u_{15} = k + 14 \left(-\frac{k}{4} \right)$$

$$u_{15} = k - \frac{7k}{2} = -\frac{5k}{2}$$

4. Given

Given

$$-\frac{5k}{2} = 90 + 2k$$

$$-5k = 180 + 4k$$

$$k = -20$$

5. Evaluate fraction

So

$$a = -20, \quad d = 5$$

$$S_{30} = \frac{30}{2}(2(-20) + 29(5))$$

$$S_{30} = 15(105) = 1575$$

Final answer

1575

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2025

Q#: 25

Marks: 5

TOPIC: Sequences

Source: Nov 2025 P2H

- 25** The first term of an arithmetic series is 10
The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

$N =$

(Total for Question 25 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 25
Sequences

1. Calculate value

$$a = 10$$

$$u_{20} = 86$$

$$10 + 19d = 86$$

$$d = 4$$

2. Use

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$5194 = \frac{N}{2}(20 + 4(N - 1))$$

$$5194 = \frac{N}{2}(4N + 16)$$

$$5194 = 2N(N + 4)$$

$$N^2 + 4N - 2597 = 0$$

$$N = \frac{-4 + 102}{2} = 49$$

Final answer

$$N = 49$$

Functions

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2020

Q#: 21

Marks: 5

Topic: **Functions**

Source: Jan 2020 P1H

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$ Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots$$

(Total for Question 21 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 21
Functions

1. Calculate value

$$f(x) = x^2 - 2x$$

$$g(x) = x + 3$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = (x + 3)^2 - 2(x + 3)$$

$$h(x) = x^2 + 4x + 3$$

$$h(x) = (x + 2)^2 - 1$$

2. Let

Let

$$y = (x + 2)^2 - 1$$

3. Find inverse function

Since the domain is $x \geq -2$, take the positive square root:

$$x + 2 = \sqrt{y + 1}$$

$$x = \sqrt{y + 1} - 2$$

Final answer

$$h^{-1}(x) = \sqrt{x + 1} - 2$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2021

Q#: 22

Marks: 3

TOPIC: **Functions**

Source: Jan 2021 P1H

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 22 is 3 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 22
Functions

1. Calculate value

$$f(x) = x^2 - 8x + 5$$

$$f(x) = (x - 4)^2 - 11$$

2. Let

Let

$$y = (x - 4)^2 - 11$$

$$y + 11 = (x - 4)^2$$

3. Find inverse function

Since the domain is $x < 4$, take the negative square root:

$$x - 4 = -\sqrt{y + 11}$$

$$x = 4 - \sqrt{y + 11}$$

Final answer

$$f^{-1}(x) = 4 - \sqrt{x + 11}$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2022

Q#: 25

Marks: 5

Topic: **Functions**

Source: Jan 2022 P2H

25 The function g is defined as

$$g: x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x: x \geq 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1}: x \mapsto \dots$

$$g^{-1}: x \mapsto \dots \quad (4)$$

(b) State the domain of g^{-1}

$$\dots \quad (1)$$

(Total for Question 25 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 25

Functions

1. Calculate value

$$g(x) = 5 + 6x - x^2$$

2. Complete the square

Complete the square:

$$g(x) = 14 - (x - 3)^2$$

3. Let

Let

$$y = 14 - (x - 3)^2$$

$$(x - 3)^2 = 14 - y$$

4. Find inverse function

The domain of g is $x \geq 3$, so take the positive square root:

$$x - 3 = \sqrt{14 - y}$$

$$x = 3 + \sqrt{14 - y}$$

5. Simplify surd

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14 - x}$$

6. Calculate statistic

The range of g is $x \leq 14$, so this is the domain of g^{-1} .**Final answer**(a) $g^{-1} : x \mapsto 3 + \sqrt{14 - x}$, (b) domain $x \leq 14$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jan 2022

Q#: 23

Marks: 4

TOPIC: **Functions**

Source: Jan 2022 P2HR

23 The functions f and g are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$ The domain of h is $\{x : x \leq 6\}$ Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 23
Functions

1. Calculate value

$$f(x) = x + 25$$

$$g(x) = x^2 - 12x$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = x^2 - 12x + 25$$

$$h(x) = (x - 6)^2 - 11$$

2. Let

Let

$$y = (x - 6)^2 - 11$$

$$y + 11 = (x - 6)^2$$

3. Find inverse function

Since the domain of h is $x < 6$, take the negative square root:

$$x - 6 = -\sqrt{y + 11}$$

$$x = 6 - \sqrt{y + 11}$$

Final answer

$$h^{-1}(x) = 6 - \sqrt{x + 11}$$

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2021

Q#: 24

Marks: 6

Topic: **Functions**

Source: May 2021 P1H

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$
(2)(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots$$

(4)

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 24
Functions

1. (a)

(a)

$$g(2) = 7(2) - 6 = 8$$

$$fg(2) = f(8)$$

$$f(8) = 5(8)^2 - 10(8) + 7$$

$$f(8) = 247$$

2. (b)

(b)

$$f(x) = 5x^2 - 10x + 7$$

$$f(x) = 5(x - 1)^2 + 2$$

3. Let

Let

$$y = 5(x - 1)^2 + 2$$

$$(x - 1)^2 = \frac{y - 2}{5}$$

4. Find inverse function

Since the domain is $x \geq 1$, take the positive square root:

$$x = 1 + \sqrt{\frac{y - 2}{5}}$$

Final answer

(a) 247, (b) $f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2024

Q#: 25

Marks: 4

Topic: **Functions**

Source: Nov 2024 P1H

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 25

Functions

1. Calculate value

$$f(x) = 2x^2 - 24x + 7$$

2. Complete the square

Complete the square:

$$f(x) = 2(x^2 - 12x) + 7$$

$$f(x) = 2[(x - 6)^2 - 36] + 7$$

$$f(x) = 2(x - 6)^2 - 65$$

3. Let

Let $y = f(x)$.

$$y = 2(x - 6)^2 - 65$$

$$y + 65 = 2(x - 6)^2$$

$$\frac{y + 65}{2} = (x - 6)^2$$

4. Simplify surd

Since $x \geq 6$, take the positive square root:

$$x - 6 = \sqrt{\frac{y + 65}{2}}$$

$$x = 6 + \sqrt{\frac{y + 65}{2}}$$

5. Replace by

Replace y by x .**Final answer**

$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$

Graphs of Functions

EA

CLASSIFIED PROBLEM

Paper: 4WM2HR

Session: May2025

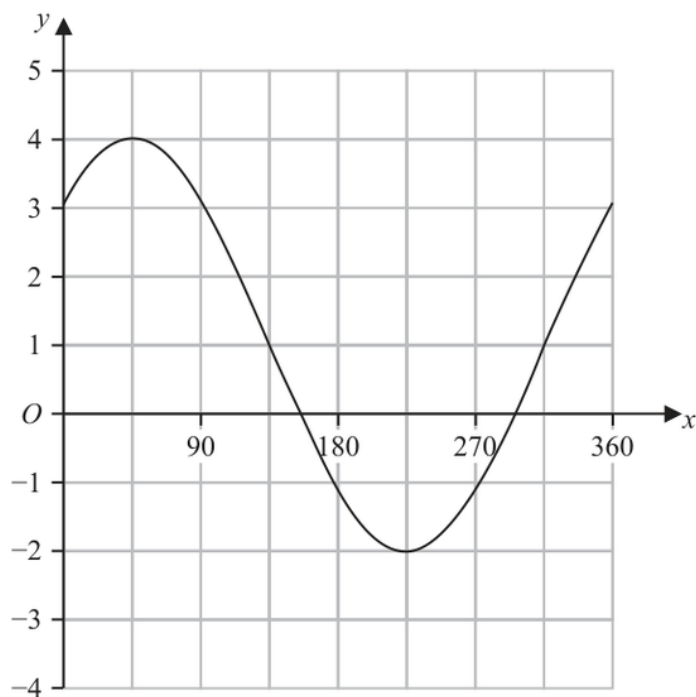
Q#: 23

Marks: 3

TOPIC: **Graphs of Functions**

Source: May 2025 4WM2HR

23 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leq x \leq 360$ is drawn on the grid below.



Find a suitable value for a , for b and for c

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 23

Graphs of Functions

1. Required

The graph of $y = a \sin (x+b)^{\circ} + c$ for $0 \leq x \leq 360$ is shown. Find suitable values for a , b , and c .

2. Read the maximum and minimum

Read the maximum and minimum values from the graph to find the **amplitude** a and **vertical shift** c :

$$a = \frac{\max - \min}{2} = \frac{4 - (-2)}{2} = 3 \quad c = \frac{\max + \min}{2} = \frac{4 + (-2)}{2} = 1$$

3. Find the phase shift b

Find the **phase shift** b . The standard sin has its peak at $x = 90^{\circ}$. From the graph, the maximum occurs at $x \approx 45^{\circ}$:

$$x + b = 90^{\circ} \implies 45^{\circ} + b = 90^{\circ} \implies b = 45^{\circ}$$

4. Verify at $x = 0$ consistent

Verify at $x = 0$: $y = 4 \sin(0 + 45)^{\circ} = 4 \times \frac{\sqrt{2}}{2} \approx 2.83$ – consistent with graph

5. Note

The period of the graph is still 360° (unchanged), so no horizontal stretch is applied.

Final answer

$$a = 3, b = 45, c = 1$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 25

Marks: 3

TOPIC: **Graphs of Functions**

Source: Nov 2025 4WM2H

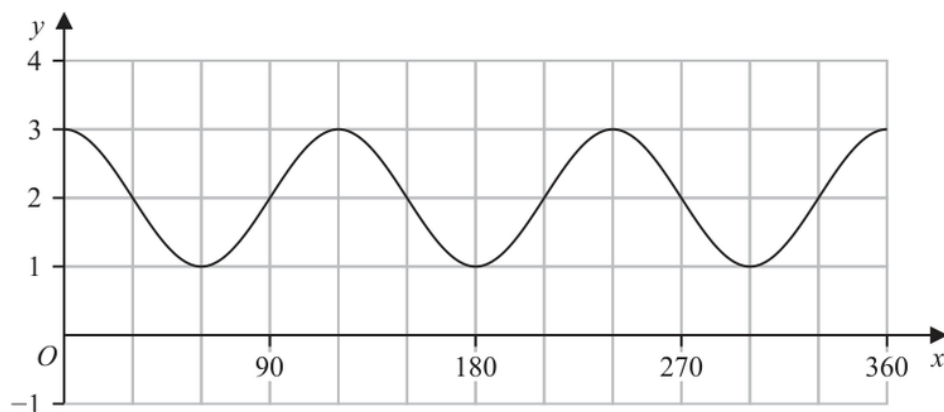
25 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(8, 1)$

- (a) Write down the coordinates of the minimum point on the curve with equation $y = f(4x)$

(.....,)
(1)

The diagram shows the graph of $y = \cos(ax) + b$ for $0 \leq x \leq 360$ 

- (b) (i) Find the value of a

$a = \dots\dots\dots$
(1)

- (ii) Find the value of b

$b = \dots\dots\dots$
(1)

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 25

Graphs of Functions

1. The transformation $y = f(4x)$

The transformation $y = f(4x)$ represents a horizontal stretch by a scale factor of $\frac{1}{4}$. The x-coordinate is multiplied by $\frac{1}{4}$ while the y-coordinate stays the same.

$$\text{New minimum} = \left(8 \times \frac{1}{4}, 1\right) = (2, 1)$$

2. Answer

(2,1)

1. Find the vertical shift b .

The standard $y = \cos(x)$ graph fluctuates between -1 and 1, centered at $y = 0$. The given graph fluctuates between a minimum of $y = 1$ and a maximum of $y = 3$. The center line is $\frac{3+1}{2} = 2$. Therefore, $b = 2$.

2. Find the frequency multiplier a .

The period of the given graph completes one full cycle between $x = 0$ and $x = 90^\circ$ (it starts at 3, dips to 1, and returns to 3 at 90°). The standard cosine period is 360° .

$$\begin{aligned}\text{Period} &= \frac{360^\circ}{a} \\ 90^\circ &= \frac{360^\circ}{a} \implies a = 4\end{aligned}$$

Final answer $a = 4, b = 2$

Transformations of Graphs

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 27

Marks: 4

Topic: Transformations of Graphs

Source: May 2025 4WM2H

- 27 A curve has equation
- $y = f(x)$

There is one maximum point on this curve.

The coordinates of this maximum point are $(6, -5)$

- (a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 4)$

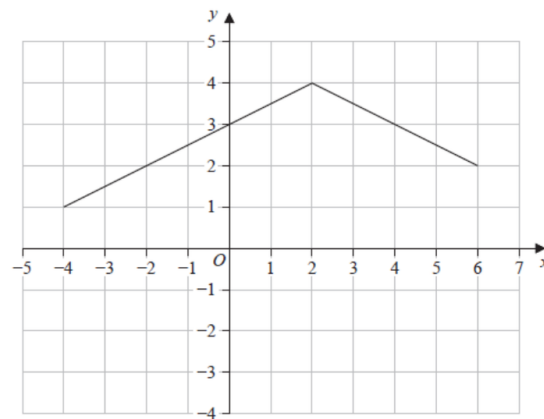
(.....,)
(1)

(ii) $y = f(3x)$

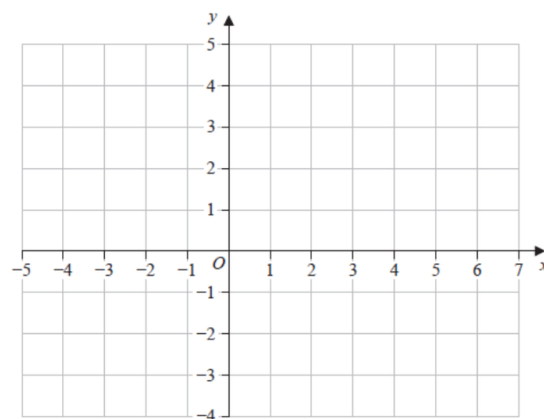
(.....,)
(1)

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Here is the graph of $y = g(x)$ 

- (b) On the grid below, draw the graph of
- $y = g(2x) - 3$



(2)

(Total for Question 27 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 27

Transformations of Graphs

1. (a)

(a)

2. Read the graph

For $y = f(x - 4)$, move the graph 4 units right:

$$(6, -5) \rightarrow (10, -5)$$

3. Divide the x -coordinateFor $y = f(3x)$, divide the x -coordinate by 3:

$$(6, -5) \rightarrow (2, -5)$$

4. Subtract from each y -coordinate(b) For $y = g(2x) - 3$, halve each x -coordinate and subtract 3 from each y -coordinate.

5. Useful transformed points are

Useful transformed points are

$$(-4, 1) \rightarrow (-2, -2), \quad (2, 4) \rightarrow (1, 1), \quad (6, 2) \rightarrow (3, -1)$$

6. Join these points with straight

Join these points with straight line segments.

Final answer

(10, -5), (2, -5), graph through (-2, -2), (1, 1), (3, -1).

Differentiation

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 22

Marks: 4

TOPIC: **Differentiation**

Source: May 2025 4WM2H

- 22 Work out the range of values of x for which the graph of $y = x^3 + 2x^2 - 10x + 5$ has a positive gradient.
Show clear algebraic working.

.....
(Total for Question 22 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 22

Differentiation

1. Evaluate fraction

$$\frac{dy}{dx} = 3x^2 + 2x - 10$$

2. Find the gradient

For a positive gradient,

$$3x^2 + 2x - 10 > 0$$

3. The roots are

The roots are

$$x = \frac{-1 \pm \sqrt{31}}{3}$$

4. Solve quadratic equation

Since the quadratic opens upwards,

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}$$

Final answer

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}.$$

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 22

Marks: 6

TOPIC: **Differentiation**

Source: Nov 2025 4WM2H

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 22

Differentiation

1. Find the gradient

$$\frac{dy}{dx} = 3x^2 - 16$$

2. Find the gradient

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9 \implies x = 3 \text{ or } x = -3$$

3. Solve inequality

$$\text{For } x = 3: y = (3)^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

4. Solve inequality

$$\text{For } x = -3: y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

5. Use trigonometry

4. Find the y-intercept (b) for each tangent ($y = 11x + b$).

Using (3,-14):

$$-14 = 11(3) + b \implies -14 = 33 + b \implies b = -47$$

Using (-3,28):

$$28 = 11(-3) + b \implies 28 = -33 + b \implies b = 61$$

Final answer

$$b = -47 \text{ and } 61$$

Transformations of Graphs

EA

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2020

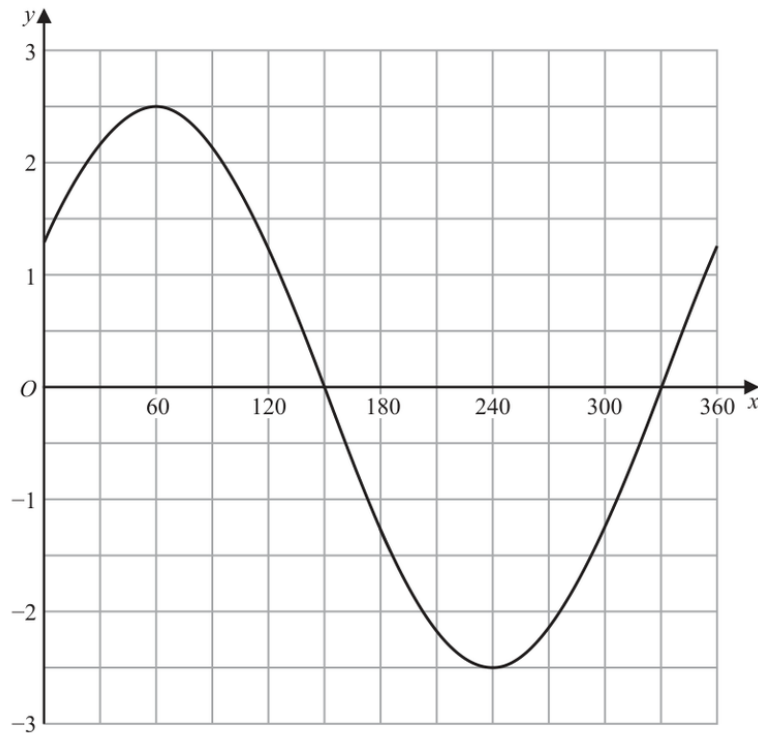
Q#: 22

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2020 P1HR

22 The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid.



(a) Find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots$
(2)

Another curve C has equation $y = f(x)$

The coordinates of the minimum point of C are $(4, 5)$

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(2x)$

(.....,)

(ii) $y = f(x) - 7$

(.....,)
(2)

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 22

Transformations of Graphs

1. Read the graphFor part (a), the graph has maximum 2.5 and minimum -2.5 , so

$$a = 2.5$$

2. Use trigonometryThe maximum occurs at $x = 60^\circ$. For $y = a \cos(x + b)^\circ$, a maximum occurs when

$$x + b = 360$$

3. Calculate value

So

$$60 + b = 360$$

$$b = 300$$

4. Calculate valueFor part (b), the minimum point of $y = f(x)$ is $(4, 5)$.**5. Calculate value**

For

$$y = f(2x)$$

6. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 5) \rightarrow (2, 5)$$

7. Calculate value

For

$$y = f(x) - 7$$

8. the coordinate decreases by 7the y -coordinate decreases by 7:

$$(4, 5) \rightarrow (4, -2)$$

Final answer

$$a = 2.5, b = 300; (2, 5) \text{ and } (4, -2)$$

CLASSIFIED PROBLEM

Paper: P2H

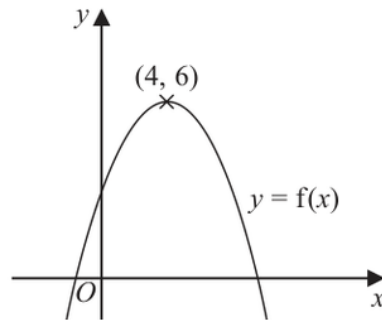
Session: Jan 2020

Q#: 21

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2020 P2H

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$ 

There is one maximum point on this curve.

The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)
(2)The equation of a curve **C** is $y = x^2 + 3x + 4$ The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ (b) Find an equation of curve **S**.*You do not need to simplify the equation.*.....
(2)**(Total for Question 21 is 4 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 21

Transformations of Graphs

1. The maximum point of isThe maximum point of $y = f(x)$ is $(4, 6)$.**2. Calculate value**

For

$$y = f(x + 4)$$

3. the graph moves 4 units

the graph moves 4 units left:

$$(4, 6) \rightarrow (0, 6)$$

4. Calculate value

For

$$y = f(2x)$$

5. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 6) \rightarrow (2, 6)$$

6. Calculate value

For part (b),

$$y = x^2 + 3x + 4$$

7. The translation vector is

The translation vector is

$$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

8. Read the graph

So the graph moves 4 units right and 6 units up.

9. Add 6Replace x by $x - 4$, then add 6:

$$y = (x - 4)^2 + 3(x - 4) + 4 + 6$$

$$y = (x - 4)^2 + 3(x - 4) + 10$$

Final answer

$(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2021

Q#: 21

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2021 P1H

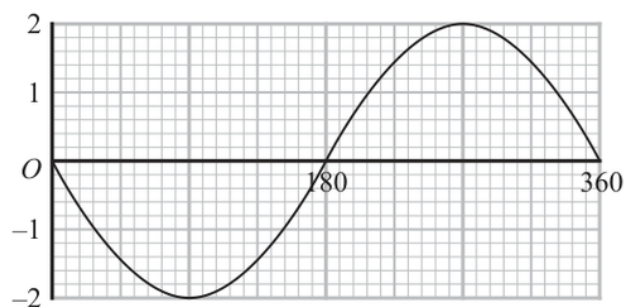
21 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(-9, 15)$

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

(.....,) (1)

(ii) $y = \frac{1}{3}f(x)$

(.....,)
(2)The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.Given that $a > 0$ and that $0 < b < 360$ (b) find the value of a and the value of b . $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 21

Transformations of Graphs

1. The minimum point of isThe minimum point of $y = f(x)$ is $(-9, 15)$.**2. Calculate value**

For

$$y = f(x + 3)$$

3. the graph moves 3 units

the graph moves 3 units left:

$$(-9, 15) \rightarrow (-12, 15)$$

4. Evaluate fraction

For

$$y = \frac{1}{3}f(x)$$

5. the coordinate is divided bythe y -coordinate is divided by 3:

$$(-9, 15) \rightarrow (-9, 5)$$

6. Use trigonometryFor part (b), the cosine graph has maximum 2 and minimum -2 , so

$$a = 2$$

7. The graph has a minimumThe graph has a minimum at $x = 90^\circ$. For $a \cos(x + b)^\circ$, a minimum occurs when

$$x + b = 180$$

8. Calculate value

So

$$90 + b = 180$$

$$b = 90$$

Final answer $(-12, 15), (-9, 5), a = 2, b = 90$

CLASSIFIED PROBLEM

Paper: P1H

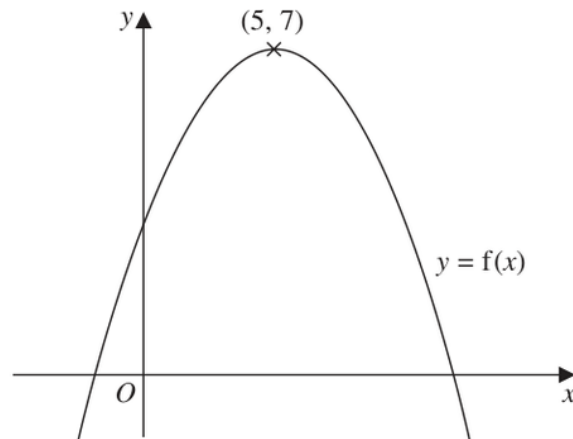
Session: Jan 2022

Q#: 23

Marks: 2

TOPIC: Transformations of Graphs

Source: Jan 2022 P1H

23 The diagram shows a sketch of the curve with equation $y = f(x)$ 

There is only one maximum point on the curve.
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

(Total for Question 23 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 23

Transformations of Graphs

1. The maximum point of isThe maximum point of $y = f(x)$ is $(5, 7)$.**2. Calculate value**

For

$$y = f(x + 9)$$

3. the graph moves 9 units

the graph moves 9 units left:

$$(5, 7) \rightarrow (-4, 7)$$

4. Calculate value

For

$$y = f(x) + 3$$

5. the graph moves 3 units

the graph moves 3 units up:

$$(5, 7) \rightarrow (5, 10)$$

Final answer $(-4, 7)$ and $(5, 10)$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2022

Q#: 20

Marks: 6

TOPIC: Transformations of Graphs

Source: Jan 2022 P1HR

20 The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

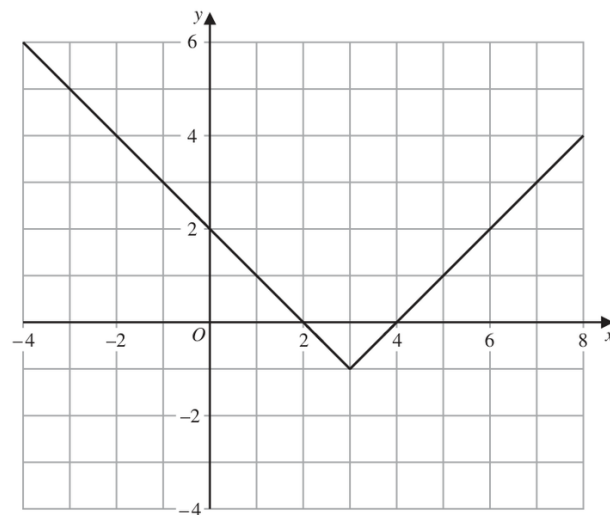
(i) $y = f(x) + 5$

(.....,)

(ii) $y = f(3x)$

(.....,)
(2)

The graph of $y = g(x)$ is shown on the grid below.



(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leq x \leq 7$

(2)

The graph of $y = h(x)$ intersects the x -axis at two points.
The coordinates of the two points are $(-1, 0)$ and $(6, 0)$

The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

(c) Find the two possible values of a

(2)

(Total for Question 20 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 20

Transformations of Graphs

1. The turning point of isThe turning point of $y = f(x)$ is $(-6, -4)$.**2. Calculate value**

For

$$y = f(x) + 5$$

3. the graph moves 5 units

the graph moves 5 units up:

$$(-6, -4) \rightarrow (-6, 1)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided bythe x -coordinate is divided by 3:

$$(-6, -4) \rightarrow (-2, -4)$$

6. Read the graphFor the sketch of $y = 2g(x)$, keep every x -coordinate the same and double every y -coordinate.**7. Useful points are**

Useful points are

$$(-1, 3) \rightarrow (-1, 6)$$

$$(2, 0) \rightarrow (2, 0)$$

$$(3, -1) \rightarrow (3, -2)$$

$$(4, 0) \rightarrow (4, 0)$$

$$(7, 3) \rightarrow (7, 6)$$

8. Solve inequalityDraw straight line segments through these points for $-1 \leq x \leq 7$.**9. Calculate value**

For part (c), $h(x)$ has roots at $x = -1$ and $x = 6$.

10. Calculate value

For $h(x + a)$ to pass through $(2, 0)$,

$$2 + a = -1 \quad \text{or} \quad 2 + a = 6$$

$$a = -3 \quad \text{or} \quad a = 4$$

Final answer

$(-6, 1)$, $(-2, -4)$, sketch through the doubled points, and $a = -3$ or $a = 4$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2022

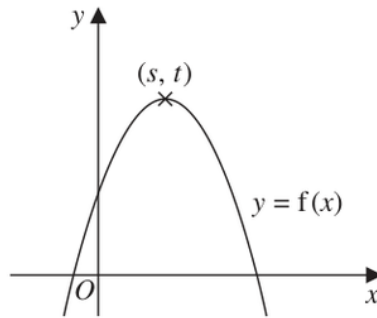
Q#: 20

Marks: 2

TOPIC: Transformations of Graphs

Source: Jun 2022 P1H

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

(..... ,)
(1)

(ii) $y = 3f(x)$

(..... ,)
(1)

(Total for Question 20 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 20

Transformations of Graphs

1. The maximum point of isThe maximum point of $y = f(x)$ is (s, t) .**2. Calculate value**

For

$$y = f(x - 2)$$

3. the graph moves 2 units

the graph moves 2 units right:

$$(s, t) \rightarrow (s + 2, t)$$

4. Calculate value

For

$$y = 3f(x)$$

5. the coordinate is multiplied bythe y -coordinate is multiplied by 3:

$$(s, t) \rightarrow (s, 3t)$$

Final answer $(s + 2, t)$ and $(s, 3t)$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 22

Marks: 4

TOPIC: Transformations of Graphs

Source: Jun 2022 P2HR

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

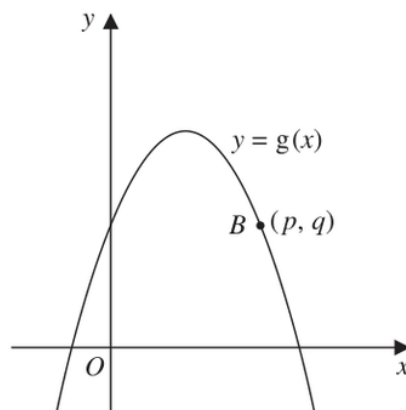
(i) $y = f(x) - 3$

(.....,)
(1)

(ii) $y = f\left(\frac{x}{2}\right)$

(.....,)
(1)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -g(x - c)$$

where c is a constant.

(.....,)
(2)

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 22

Transformations of Graphs

1. Point lies onPoint $A = (-3, 2)$ lies on $y = f(x)$.**2. Calculate value**

For

$$y = f(x) - 3$$

3. the graph moves 3 units

the graph moves 3 units down:

$$(-3, 2) \rightarrow (-3, -1)$$

4. Evaluate fraction

For

$$y = f\left(\frac{x}{2}\right)$$

5. the coordinate is multiplied bythe x -coordinate is multiplied by 2:

$$(-3, 2) \rightarrow (-6, 2)$$

6. Calculate valueFor part (b), point $B = (p, q)$ lies on $y = g(x)$.**7. Calculate value**

For

$$y = -g(x - c)$$

8. Read the graph $x - c$ moves the graph c units right, and the minus sign reflects the graph in the x -axis.**9. Apply transformation**

So

$$(p, q) \rightarrow (p + c, -q)$$

Final answer $(-3, -1)$, $(-6, 2)$, and $(p + c, -q)$

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 23

Marks: 2

Topic: **Transformations of Graphs**

Source: Jun 2024 P2HR

23 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.....,)
(1)

(ii) $y = f(3x)$

(.....,)
(1)**(Total for Question 23 is 2 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 23

Transformations of Graphs

1. The minimum point of isThe minimum point of $y = f(x)$ is $(6, -3)$.**2. Calculate value**

For

$$y = f(x) + 10$$

3. the graph moves 10 units

the graph moves 10 units up:

$$(6, -3) \rightarrow (6, 7)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided bythe x -coordinate is divided by 3:

$$(6, -3) \rightarrow (2, -3)$$

Final answer $(6, 7)$ and $(2, -3)$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: Transformations of Graphs

Source: Jun 2025 P1HR

- 21** The curve with equation $y = f(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve
with equation $y = f(3x)$

(.....,)
(1)

The curve **C** with equation $y = g(x)$ is transformed to give the curve **S** with
equation $y = g(x + a) + b$

The point $(4, -5)$ on **C** is mapped to the point $(1, -16)$ on **S**

- (b) Write down the value of a and the value of b

$a = \dots\dots\dots$

$b = \dots\dots\dots$
(2)

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 21

Transformations of Graphs

1. The turning point of isThe turning point of $y = f(x)$ is $(-6, 9)$.**2. Calculate value**

For

$$y = f(3x)$$

3. the coordinate is divided bythe x -coordinate is divided by 3:

$$(-6, 9) \rightarrow (-2, 9)$$

4. Calculate valueFor part (b), a point $(4, -5)$ on $y = g(x)$ maps to $(1, -16)$ on

$$y = g(x + a) + b$$

5. Calculate valueFor the x -coordinate:

$$1 + a = 4$$

$$a = 3$$

6. Calculate valueFor the y -coordinate:

$$-5 + b = -16$$

$$b = -11$$

Final answer $(-2, 9)$, $a = 3$, $b = -11$

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2023

Q#: 21

Marks: 2

TOPIC: **Transformations of Graphs**

Source: May 2023 P1H

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.....,)
(1)

(b) $y = f(3x)$

(.....,)
(1)**(Total for Question 21 is 2 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 21

Transformations of Graphs

1. The turning point of isThe turning point of $y = f(x)$ is $(6, 5)$.**2. Calculate value**

For

$$y = f(x - 4)$$

3. the graph moves 4 units

the graph moves 4 units right:

$$(6, 5) \rightarrow (10, 5)$$

4. Calculate value

For

$$y = f(3x)$$

5. the coordinate is divided bythe x -coordinate is divided by 3:

$$(6, 5) \rightarrow (2, 5)$$

Final answer $(10, 5)$ and $(2, 5)$

CLASSIFIED PROBLEM

Paper: P1HR

Session: May 2023

Q#: 20

Marks: 7

TOPIC: Transformations of Graphs

Source: May 2023 P1HR

20 Curve **C** has equation $y = f(x)$ The graph of curve **C** has one maximum point.

The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.....,)
(1)

(ii) $y = f(x) - 7$

(.....,)
(1)

(iii) $y = f(-x)$

(.....,)
(1)Curve **L** has equation $y = x^2 + 7x + 20$ Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (b) Find an equation for **S**Give your answer in the form $y = ax^2 + bx + c$ $y = \dots\dots\dots$
(4)

(Total for Question 20 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 20

Transformations of Graphs

1. The maximum point of isThe maximum point of $y = f(x)$ is $(3, 5)$.**2. Calculate value**

For

$$y = 2f(x)$$

3. the coordinate is multiplied bythe y -coordinate is multiplied by 2:

$$(3, 5) \rightarrow (3, 10)$$

4. Calculate value

For

$$y = f(x) - 7$$

5. the graph moves 7 units

the graph moves 7 units down:

$$(3, 5) \rightarrow (3, -2)$$

6. Calculate value

For

$$y = f(-x)$$

7. the graph reflects in thethe graph reflects in the y -axis:

$$(3, 5) \rightarrow (-3, 5)$$

8. Calculate value

For part (b),

$$y = x^2 + 7x + 20$$

9. The translation vector is

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

10. Read the graph

So the graph moves 2 units right.

11. Replace by

Replace x by $x - 2$:

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

Final answer

$(3, 10)$, $(3, -2)$, $(-3, 5)$, and $y = x^2 + 3x + 10$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2024

Q#: 21

Marks: 2

TOPIC: Transformations of Graphs

Source: May 2024 P1H

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(..... ,)
(1)

(ii) $y = f(x) - 6$

(..... ,)
(1)

(Total for Question 21 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 21

Transformations of Graphs

1. The minimum point of isThe minimum point of $y = f(x)$ is $(5, -4)$.**2. Calculate value**

For

$$y = f(x + 7)$$

3. the graph moves 7 units

the graph moves 7 units left:

$$(5, -4) \rightarrow (-2, -4)$$

4. Calculate value

For

$$y = f(x) - 6$$

5. the graph moves 6 units

the graph moves 6 units down:

$$(5, -4) \rightarrow (5, -10)$$

Final answer $(-2, -4)$ and $(5, -10)$

CLASSIFIED PROBLEM

Paper: P1H

Session: MayNov 2020

Q#: 25

Marks: 3

TOPIC: Transformations of Graphs

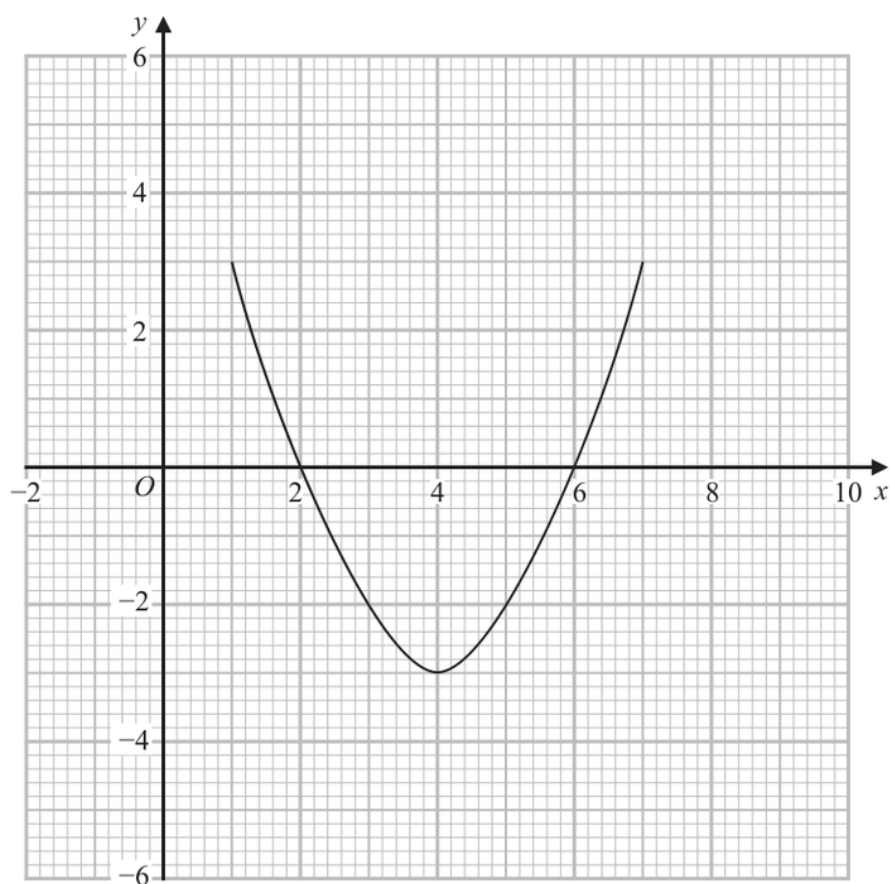
Source: May-Nov 2020 P1H

- 25** The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 25

Transformations of Graphs

1. Apply transformation

For part (a),

$$y = g(x) \rightarrow y = -g(x)$$

2. The coordinates stay the sameThe x -coordinates stay the same and the y -coordinates change sign.**3. Apply transformation**So the transformation is a reflection in the x -axis.**4. Calculate value**

For part (b),

$$y = 2f(x - 1)$$

5. Read the graphFirst, $x - 1$ moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.**6. Use points from the original**

Use points from the original graph:

$$(1, 3) \rightarrow (2, 6)$$

$$(2, 0) \rightarrow (3, 0)$$

$$(4, -3) \rightarrow (5, -6)$$

$$(6, 0) \rightarrow (7, 0)$$

$$(7, 3) \rightarrow (8, 6)$$

7. Apply transformation

Draw the transformed parabola through these points.

Final answerreflection in the x -axis; transformed graph through $(2, 6)$, $(3, 0)$, $(5, -6)$, $(7, 0)$, $(8, 6)$

CLASSIFIED PROBLEM

Paper: P1HR

Session: MayNov 2020

Q#: 20

Marks: 2

TOPIC: Transformations of Graphs

Source: May-Nov 2020 P1HR

20 A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(.....,)

(ii) $y = f(2x)$

(.....,)

(Total for Question 20 is 2 marks)

E A

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 20
Transformations of Graphs

1. The maximum point of is

The maximum point of $y = f(x)$ is $(-3, 4)$.

2. Calculate value

For

$$y = f(x) - 6$$

3. the graph moves 6 units

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

4. Calculate value

For

$$y = f(2x)$$

5. the coordinate is divided by

the x -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

Final answer

$(-3, -2)$ and $\left(-\frac{3}{2}, 4\right)$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2021

Q#: 21

Marks: 6

TOPIC: Transformations of Graphs

Source: Nov 2021 P2H

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

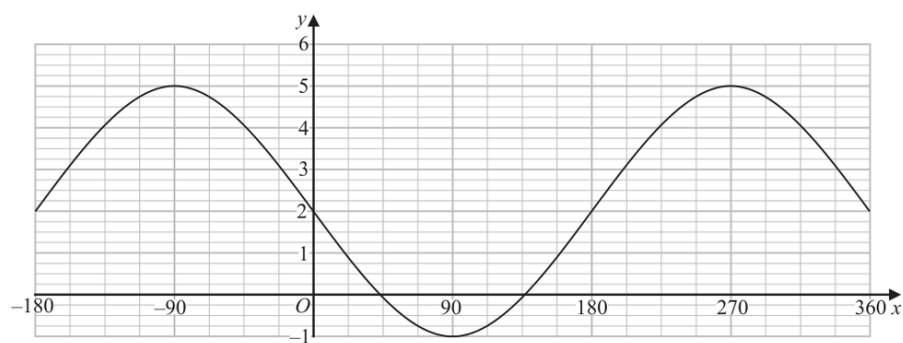
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(3)

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P2H | Question 21

Transformations of Graphs

1. Calculate value

$$f(x) = 9 - 3(x + 2)^2$$

2. This is in completed square

This is in completed square form, so the maximum point is

$$(-2, 9)$$

3. Use matrices

For the translation

$$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

4. the graph moves 4 units

the graph moves 4 units right. Replace x by $x - 4$:

$$y = 9 - 3((x - 4) + 2)^2$$

$$y = 9 - 3(x - 2)^2$$

5. The curve has equation

The curve T has equation

$$y = 3(x + 2)^2 - 9$$

6. This is

This is

$$y = -\left(9 - 3(x + 2)^2\right)$$

7. Apply transformation

So C is reflected in the x -axis.

8. Use trigonometry

For the cosine graph, the maximum is 5 and the minimum is -1 .

$$a = \frac{5 - (-1)}{2} = 3$$

$$c = \frac{5 + (-1)}{2} = 2$$

9. Use trigonometry

For

$$y = a \cos(x - b)^\circ + c$$

10. Read the graph

a maximum occurs when $x - b = 0$. The graph has a maximum at $x = 270^\circ$, so

$$b = 270$$

Final answer

$A = (-2, 9)$, $y = 9 - 3(x - 2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2023

Q#: 24

Marks: 4

TOPIC: Transformations of Graphs

Source: Nov 2023 P1H

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

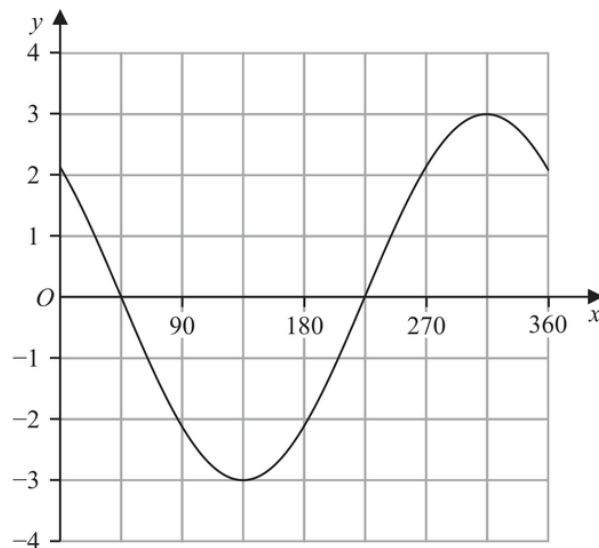
(a) Write down the value of a and the value of b

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2)

The graph of $y = p \cos(x+q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots$$

(2)

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 24

Transformations of Graphs

1. Read the graph

For part (a),

$$f(x) = 5x^2 + 9x + 2$$

$$g(x) = 5(x + 4)^2 + 9(x + 4) + 8$$

2. Read the graph

Since

$$f(x + 4) = 5(x + 4)^2 + 9(x + 4) + 2$$

$$g(x) = f(x + 4) + 6$$

3. Read the graph

So the graph has moved 4 units left and 6 units up.

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$$

4. Read the graph

For part (b), the graph has maximum value 3 and minimum value -3 , so

$$p = 3$$

5. The minimum occurs when

The minimum occurs when

$$x + q = 180$$

6. Read the graph

From the graph, the minimum is at $x = 135$, so

$$135 + q = 180$$

$$q = 45$$

Final answer

$$a = -4, b = 6, p = 3, q = 45$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2025

Q#: 24

Marks: 3

TOPIC: Transformations of Graphs

Source: Nov 2025 P1H

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)
(1)

(ii) $y = f(x) + 7$

(..... ,)
(1)

(iii) $y = f(2x)$

(..... ,)
(1)

(Total for Question 24 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 24

Transformations of Graphs

1. The turning point of isThe turning point of $y = f(x)$ is $(4, 3)$.**2. Calculate value**

For

$$y = f(x + 5)$$

3. the graph moves 5 units

the graph moves 5 units left:

$$(4, 3) \rightarrow (-1, 3)$$

4. Calculate value

For

$$y = f(x) + 7$$

5. the graph moves 7 units

the graph moves 7 units up:

$$(4, 3) \rightarrow (4, 10)$$

6. Calculate value

For

$$y = f(2x)$$

7. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 3) \rightarrow (2, 3)$$

Final answer $(-1, 3)$, $(4, 10)$, and $(2, 3)$

Differentiation

EA

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2020

Q#: 23

Marks: 6

TOPIC: **Differentiation**

Source: Jan 2020 P1HR

23 A particle moves along a straight line.The fixed point O lies on this line.The displacement of the particle from O at time t seconds, $t \geq 0$, is s metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time T seconds the velocity of P is V m/s where $V \geq -5$ Find an expression for T in terms of V .Give your expression in the form $\frac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found. $T = \dots\dots\dots$ **(Total for Question 23 is 6 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 23

Differentiation

1. Calculate value

$$s = t^3 + 4t^2 - 5t + 7$$

2. Evaluate fraction

Velocity is the derivative of displacement:

$$v = \frac{ds}{dt} = 3t^2 + 8t - 5$$

3. Calculate value

At time T , the velocity is V , so

$$V = 3T^2 + 8T - 5$$

$$3T^2 + 8T - (V + 5) = 0$$

4. Use the quadratic formula

Use the quadratic formula:

$$T = \frac{-8 \pm \sqrt{8^2 - 4(3)(-(V+5))}}{2(3)}$$

$$T = \frac{-8 \pm \sqrt{64 + 12V + 60}}{6}$$

$$T = \frac{-8 \pm \sqrt{124 + 12V}}{6}$$

$$T = \frac{-4 \pm \sqrt{31 + 3V}}{3}$$

5. Simplify surd

Since $T \geq 0$, take the positive root:

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

Final answer

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2021

Q#: 20

Marks: 5

TOPIC: Differentiation

Source: Jan 2021 P1HR

- 20** A particle P is moving along a straight line.
The fixed point O lies on the line.

At time t seconds ($t \geq 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P .

..... m/s

(Total for Question 20 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 20

Differentiation

1. Read the graph

$$s = t^3 - 9t^2 + 33t - 6$$

2. Read the graph

Velocity is the rate of change of displacement:

$$v = \frac{ds}{dt}$$

$$v = 3t^2 - 18t + 33$$

3. Complete the square

Complete the square:

$$v = 3(t^2 - 6t + 11)$$

$$v = 3[(t - 3)^2 + 2]$$

$$v = 3(t - 3)^2 + 6$$

4. Read the graph

Since $(t - 3)^2 \geq 0$, the minimum value of v is

$$6$$

5. Read the graph

The velocity is always positive, so the minimum speed is also 6.

Final answer

6 m/s

CLASSIFIED PROBLEM

Paper: P1H

Session: Jan 2022

Q#: 24

Marks: 6

TOPIC: Differentiation

Source: Jan 2022 P1H

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 24

Differentiation

1. Calculate value

$$y = ax^3 + bx^2 - 12x + 6$$

2. Point lies on the curve

Point $A = (2, -6)$ lies on the curve:

$$-6 = 8a + 4b - 24 + 6$$

$$8a + 4b = 12$$

$$2a + b = 3$$

3. Differentiate

Differentiate:

$$\frac{dy}{dx} = 3ax^2 + 2bx - 12$$

4. The gradient at is 16

The gradient at $x = 2$ is 16:

$$16 = 12a + 4b - 12$$

$$12a + 4b = 28$$

$$3a + b = 7$$

5. Solve

Now solve

$$2a + b = 3$$

$$3a + b = 7$$

6. Subtracting gives

Subtracting gives

$$a = 4$$

7. Calculate value

Then

$$2(4) + b = 3$$

$$b = -5$$

8. Calculate value

For $x = 3$,

$$y = 4(3)^3 - 5(3)^2 - 12(3) + 6$$

$$y = 108 - 45 - 36 + 6$$

$$y = 33$$

Final answer

$$y = 33$$

EA

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2022

Q#: 23

Marks: 6

TOPIC: **Differentiation**

Source: Jan 2022 P1HR

- 23** Two particles, P and Q , move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 23

Differentiation

1. Particle

Particle P :

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t - 5)(t - 1)$$

2. Particle

Particle Q :

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t - 2)$$

3. Differentiate

For $t > 1$:

4. moves in the negative direction

- P moves in the negative direction when $1 < t < \frac{5}{3}$.
- P moves in the positive direction when $t > \frac{5}{3}$.
- Q moves in the negative direction when $1 < t < 2$.
- Q moves in the positive direction when $t > 2$.

5. Differentiate

So both particles move in the same direction when both velocities have the same sign.

Final answer

$$1 < t < \frac{5}{3} \text{ or } t > 2$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2022

Q#: 24

Marks: 5

TOPIC: Differentiation

Source: Jan 2022 P2H

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

(Total for Question 24 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 24

Differentiation

1. Split the ratio

$$x = 4t^3 - 27t + 8$$

2. Velocity is

Velocity is

$$\frac{dx}{dt} = 12t^2 - 27$$

3. The direction of motion reverses

The direction of motion reverses when velocity is zero:

$$12t^2 - 27 = 0$$

$$12t^2 = 27$$

$$t^2 = \frac{9}{4}$$

4. Split the ratio

Since $t \geq 0$, $t = \frac{3}{2}$.

5. Acceleration is

Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

6. Split the ratio

At $t = \frac{3}{2}$,

$$a = 24 \left(\frac{3}{2} \right) = 36$$

Final answer

$$a = 36$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2022

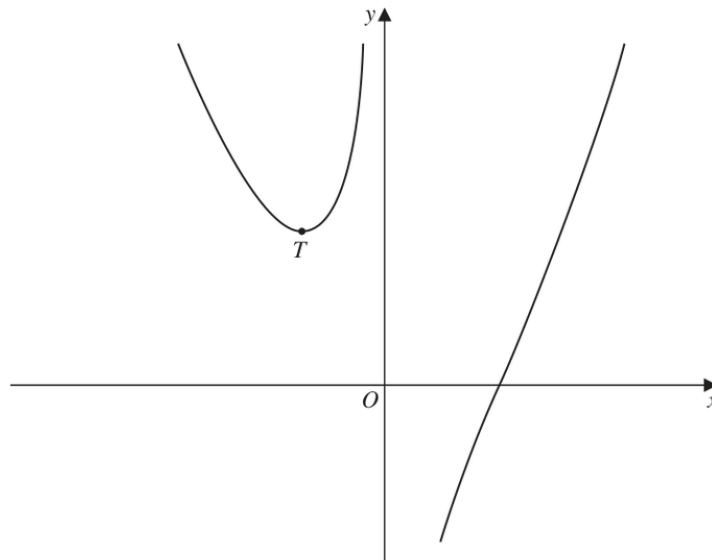
Q#: 22

Marks: 7

TOPIC: Differentiation

Source: Jun 2022 P1HR

- 22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

Diagram **NOT**
accurately drawn

For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

- (a) find the value of p

$$p = \dots\dots\dots$$

(4)

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

- (b) Find the value of k

$$k = \dots\dots\dots$$

(3)

(Total for Question 22 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 22

Differentiation

1. Find the gradient

For part (a),

$$y = x^2 - \frac{p}{x}$$

$$\frac{dy}{dx} = 2x + \frac{p}{x^2}$$

2. Find the gradientAt the minimum point, $x = -3$ and the gradient is 0.

$$2(-3) + \frac{p}{(-3)^2} = 0$$

$$-6 + \frac{p}{9} = 0$$

$$p = 54$$

3. Find the gradient

For part (b),

$$y = x^2 - \frac{16}{x}$$

$$\frac{dy}{dx} = 2x + \frac{16}{x^2}$$

4. Find the gradient

At a horizontal tangent,

$$2x + \frac{16}{x^2} = 0$$

$$2x^3 + 16 = 0$$

$$x^3 = -8$$

$$x = -2$$

5. Find the gradientNow substitute $x = -2$ into the curve:

$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

Final answer

$$p = 54, k = 12$$

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: Differentiation

Source: Jun 2025 P2HR

22 The diagram shows a solid cuboid.

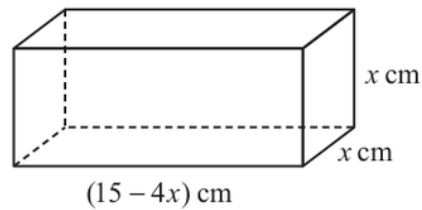


Diagram **NOT**
accurately drawn

The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

(Total for Question 22 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 22

Differentiation

1. The cuboid has dimensions

The cuboid has dimensions

$$x, \quad x, \quad 15 - 4x$$

2. Calculate value

So

$$V = x^2(15 - 4x)$$

$$V = 15x^2 - 4x^3$$

3. Differentiate

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

4. Calculate value

At a maximum,

$$30x - 12x^2 = 0$$

$$6x(5 - 2x) = 0$$

5. The useful value is

The useful value is

$$x = 2.5$$

6. Calculate volume

Now substitute into the volume:

$$V = (2.5)^2(15 - 4(2.5))$$

$$V = 6.25 \times 5$$

$$V = 31.25$$

Final answer

$$31.25 \text{ cm}^3$$

CLASSIFIED PROBLEM

Paper: P2H

Session: May 2021

Q#: 21

Marks: 5

TOPIC: Differentiation

Source: May 2021 P2H

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 21

Differentiation

1. Evaluate fraction

$$y = kx^2 + \frac{16}{x}$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 2kx - \frac{16}{x^2}$$

3. Find the gradient

At the stationary point, $x = \frac{2}{3}$, so the gradient is 0:

$$2k \left(\frac{2}{3} \right) - \frac{16}{\left(\frac{2}{3} \right)^2} = 0$$

$$\frac{4k}{3} - \frac{16}{\frac{4}{9}} = 0$$

$$\frac{4k}{3} - 36 = 0$$

$$\frac{4k}{3} = 36$$

$$k = 27$$

4. Find the -coordinate

Now find a , the y -coordinate when $x = \frac{2}{3}$:

$$a = 27 \left(\frac{2}{3} \right)^2 + \frac{16}{\frac{2}{3}}$$

$$a = 27 \cdot \frac{4}{9} + 24$$

$$a = 12 + 24 = 36$$

Final answer

$$a = 36$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: May 2023

Q#: 21

Marks: 6

TOPIC: Differentiation

Source: May 2023 P1HR

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

(4)

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 21

Differentiation

1. (a)

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

2. (b) The curve has a

(b) The curve has a positive gradient when

$$3x^2 - 4x - 9 > 0$$

3. Solve

Solve

$$3x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 108}}{6}$$

$$x = \frac{2 \pm \sqrt{31}}{3}$$

4. Differentiate

To 3 significant figures, the roots are

$$x = -1.19, \quad x = 2.52$$

5. Solve quadratic equation

Since the quadratic opens upwards, it is positive outside the roots.

Final answer

$$(a) \frac{dy}{dx} = 3x^2 - 4x - 9, (b) x < -1.19 \text{ or } x > 2.52$$

CLASSIFIED PROBLEM

Paper: P1H

Session: MayNov 2020

Q#: 23

Marks: 4

TOPIC: Differentiation

Source: May-Nov 2020 P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 23

Differentiation

1. Differentiate

$$y = px^3 - mx$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

3. The gradient is negative when

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

4. Simplify surd

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

Final answer

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2024

Q#: 20

Marks: 4

TOPIC: **Differentiation**

Source: Nov 2024 P1H

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

(Total for Question 20 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 20

Differentiation

1. Find the gradient

$$y = 2x^4 - 64x$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

3. Find the gradient

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

4. Find the coordinate

Find the y -coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

5. The tangent at a minimum

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

Final answer

$$y = -96$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2025

Q#: 22

Marks: 6

TOPIC: **Differentiation**

Source: Nov 2025 P2H

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 22

Differentiation

1. Differentiate

$$y = x^3 - 16x + 7$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 3x^2 - 16$$

3. The gradient of the tangent

The gradient of the tangent is 11, so

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9$$

$$x = 3 \quad \text{or} \quad x = -3$$

4. Differentiate

When $x = 3$:

$$y = 3^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

5. Use trigonometry

The tangent has equation $y = 11x + b$, so

$$-14 = 11(3) + b$$

$$b = -47$$

6. Differentiate

When $x = -3$:

$$y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

$$28 = 11(-3) + b$$

$$b = 61$$

Final answer

$$b = -47 \text{ or } b = 61$$

EA

Angles in Polygons & Parallel Lines

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Jun 2022

Q#: 23

Marks: 6

TOPIC: Angles in Polygons & Parallel Lines

Source: Jun 2022 P1H

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n
Show clear algebraic working.

Question 23 continued $n = \dots\dots\dots$

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 23

Angles in Polygons & Parallel Lines

1. The interior angles form an

The interior angles form an arithmetic sequence:

$$177, 175, 173, 171, 169, \dots$$

2. Calculate value

So

$$a = 177, \quad d = -2$$

3. Use trigonometryFor an n -sided polygon, the sum of the interior angles is

$$180(n - 2)$$

4. The sum of the arithmetic

The sum of the arithmetic sequence is

$$\begin{aligned} & \frac{n}{2} (2(177) + (n - 1)(-2)) \\ &= \frac{n}{2} (354 - 2n + 2) \\ &= \frac{n}{2} (356 - 2n) \\ &= n(178 - n) \end{aligned}$$

5. Set this equal to the

Set this equal to the polygon angle sum:

$$n(178 - n) = 180(n - 2)$$

$$178n - n^2 = 180n - 360$$

$$n^2 + 2n - 360 = 0$$

6. Factorise

Factorise:

$$(n + 20)(n - 18) = 0$$

7. Solve inequality

Since $n > 5$,

$$n = 18$$

Final answer

$$n = 18$$

EA

Bearings, Scale Drawing & Constructions

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2024

Q#: 24

Marks: 6

TOPIC: Bearings, Scale Drawing & Constructions

Source: Nov 2024 P2H

24 A , B and C are three points on horizontal ground.

$AB = 8.4$ metres $BC = 9.2$ metres

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

o

(Total for Question 24 is 6 marks)

ELA

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P2H | Question 24

Bearings, Scale Drawing & Constructions

1. Resolve each journey into east

Resolve each journey into east and north components.

2. Use trigonometry

From A to B :

$$\text{east} = 8.4 \sin 67^\circ = 7.7322 \dots$$

$$\text{north} = 8.4 \cos 67^\circ = 3.2821 \dots$$

3. Use trigonometry

From B to C :

$$\text{east} = 9.2 \sin 129^\circ = 7.1497 \dots$$

$$\text{north} = 9.2 \cos 129^\circ = -5.7897 \dots$$

4. Calculate value

So from A to C :

$$\text{east} = 14.8820 \dots$$

$$\text{north} = -2.5076 \dots$$

5. Calculate value

Therefore from C to A :

$$\text{east} = -14.8820 \dots, \quad \text{north} = 2.5076 \dots$$

6. Use trigonometry

This direction is west of north.

$$\theta = \tan^{-1} \left(\frac{14.8820 \dots}{2.5076 \dots} \right) = 80.4355 \dots^\circ$$

7. Use bearings

Bearing of A from C :

$$360^\circ - 80.4355 \dots^\circ = 279.5644 \dots^\circ$$

Final answer

280°

Circle Theorems

EA

CLASSIFIED PROBLEM

Paper: P2HR

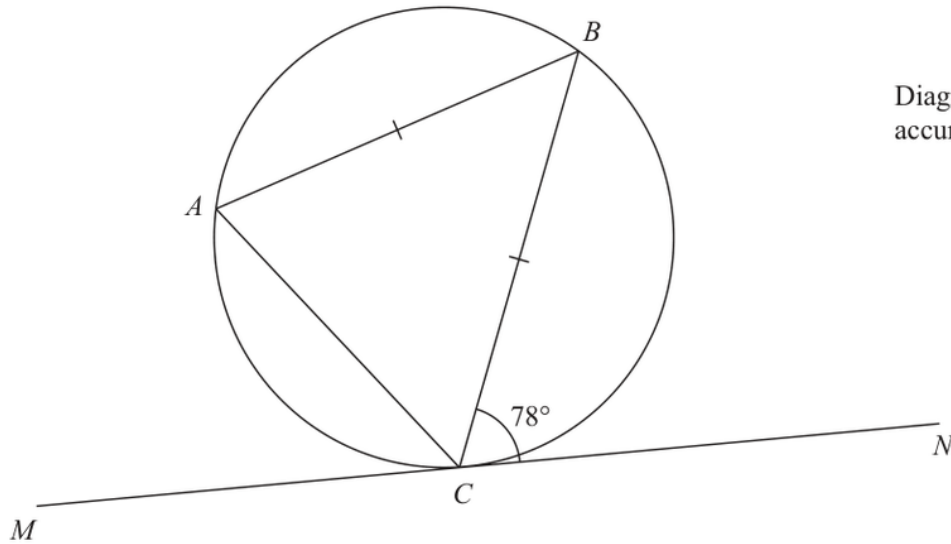
Session: Jun 2023

Q#: 20

Marks: 2

TOPIC: Circle Theorems

Source: Jun 2023 P2HR

20 A , B and C are points on a circle.Diagram **NOT**
accurately drawn MN is the tangent to the circle at C $AB = CB$ Angle $BCN = 78^\circ$ Find the size of angle ABC

o

(Total for Question 20 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 20

Circle Theorems

1. The angle between the tangent

The angle between the tangent CN and the chord CB is 78° .

2. Use trigonometry

By the alternate segment theorem,

$$\angle BAC = 78^\circ$$

3. Calculate value

Also,

$$AB = CB$$

4. Use trigonometry

so triangle ABC is isosceles and the base angles are equal:

$$\angle BAC = \angle ACB = 78^\circ$$

5. Use trigonometry

Using the angle sum of a triangle:

$$\angle ABC = 180^\circ - 78^\circ - 78^\circ$$

$$\angle ABC = 24^\circ$$

Final answer

24°

CLASSIFIED PROBLEM

Paper: P1HR

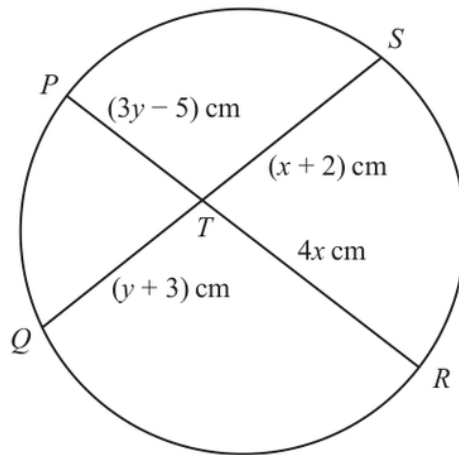
Session: Jun 2025

Q#: 23

Marks: 5

TOPIC: Circle Theorems

Source: Jun 2025 P1HR

23Diagram **NOT**
accurately drawn PTR and QTS are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for y in terms of x

$$y = \dots\dots\dots$$

Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 23

Circle Theorems

1. Use chord theorem

For two intersecting chords in a circle,

$$PT \times TR = QT \times TS$$

2. Substitute the given lengths

Substitute the given lengths:

$$(3y - 5)(4x) = (y + 3)(x + 2)$$

3. Expand

Expand:

$$12xy - 20x = xy + 2y + 3x + 6$$

4. Collect the terms on oneCollect the y terms on one side:

$$12xy - xy - 2y = 20x + 3x + 6$$

$$11xy - 2y = 23x + 6$$

5. Factorise

Factorise:

$$y(11x - 2) = 23x + 6$$

$$y = \frac{23x + 6}{11x - 2}$$

Final answer

$$y = \frac{23x + 6}{11x - 2}$$

Volume & Surface Area

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 23

Marks: 3

TOPIC: **Volume & Surface Area**

Source: May 2025 4WM2H

- 23 The diagram shows a cuboid.

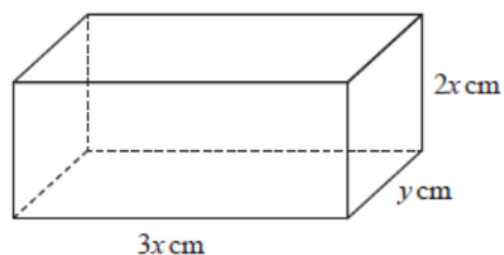


Diagram NOT
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A \text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 23
Volume & Surface Area

1. Evaluate fraction

$$3x \times 2x \times y = 1014$$

$$6x^2y = 1014$$

$$y = \frac{169}{x^2}$$

2. Surface area

Surface area:

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$= 12x^2 + 10xy$$

$$= 12x^2 + 10x \left(\frac{169}{x^2} \right)$$

$$= 12x^2 + \frac{1690}{x}$$

Final answer

$$A = 12x^2 + \frac{1690}{x}.$$

CLASSIFIED PROBLEM

Paper: 4WM2HR

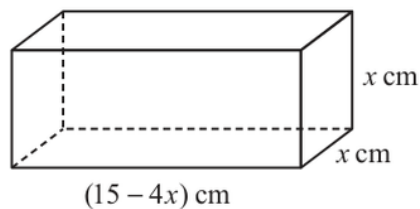
Session: May2025

Q#: 21

Marks: 5

TOPIC: **Volume & Surface Area**

Source: May 2025 4WM2HR

21 The diagram shows a solid cuboid.Diagram **NOT**
accurately drawnThe volume of the cuboid is $V \text{ cm}^3$ Find the maximum value of V

(Total for Question 21 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 21

Volume & Surface Area

1. Required

Cuboid has a square cross-section of side x cm and length $(15-4x)$ cm. Find the maximum value of the volume V .

2. Write V as a function

Write V as a function of x :

$$V = x \cdot x \cdot (15-4x) = x^2(15-4x) = 15x^2 - 4x^3$$

3. Differentiate

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

4. Set $\frac{dV}{dx} = 0$ gives

Set $\frac{dV}{dx} = 0$:

$$30x - 12x^2 = 0 \implies 6x(5 - 2x) = 0 \implies x = 0 \text{ or } x = \frac{5}{2} = 2.5$$

($x = 0$ gives zero volume, so the meaningful solution is $x = 2.5$.)

5. Second derivative test to confirm

Second derivative test to confirm maximum:

$$\frac{d^2V}{dx^2} = 30 - 24x \Bigg|_{x=2.5} = 30 - 60 = -30 < 0 \quad \checkmark \text{ (maximum)}$$

6. Calculate $V_{\max}$

Calculate $V_{\max}$:

$$V_{\max} = (2.5)^2 \times (15-4 \times 2.5) = 6.25 \times 5 = 31.25 \text{ cm}^3$$

7. Note

Calculus optimisation: always verify the nature of the stationary point using $\frac{d^2V}{dx^2}$. Also check $x > 0$ and $15 - 4x > 0$ (i.e. $x < 3.75$) to ensure a valid cuboid.

Final answer

Maximum volume $V = 31.25 \text{ cm}^3$

CLASSIFIED PROBLEM

Paper: 4WM2HR

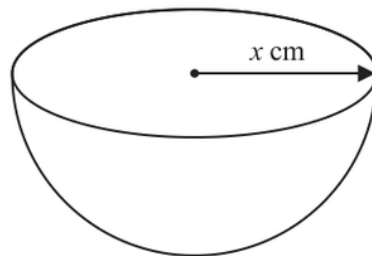
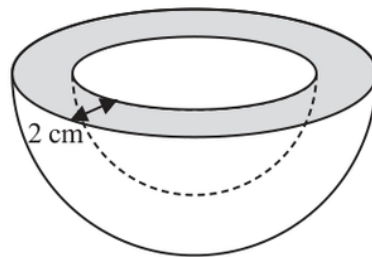
Session: May2025

Q#: 25

Marks: 5

TOPIC: **Volume & Surface Area**

Source: May 2025 4WM2HR

25 The diagram shows a solid hemisphere, H Diagram **NOT**
accurately drawnThe radius of H is x cmThe volume of H is 6174π cm³A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cmDiagram **NOT**
accurately drawnWork out the **total** surface area of the bowl.
Give your answer in terms of π

..... cm²**(Total for Question 25 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 25

Volume & Surface Area

1. Required

Hemisphere H has radius x cm and volume 6174π cm³. A bowl is made by removing an inner hemisphere, leaving 2 cm thickness. Find total surface area in terms of π .

2. Calculate volume

Find outer radius x . Volume of hemisphere = $\frac{2}{3}\pi r^3$:

$$\frac{2}{3}\pi x^3 = 6174\pi \implies x^3 = \frac{6174 \times 3}{2} = 9261 \implies x = \sqrt[3]{9261} = 21 \text{ cm}$$

3. Find inner radius r inner

Find inner radius:

$$r_{\text{inner}} = 21 - 2 = 19 \text{ cm}$$

4. Add 1

Three surfaces to add:

1. Outer curved surface (half-sphere, radius 21):

$$A_1 = 2\pi(21)^2 = 2\pi \times 441 = 882\pi$$

2. Inner curved surface (half-sphere, radius 19):

$$A_2 = 2\pi(19)^2 = 2\pi \times 361 = 722\pi$$

3. Flat annular ring at the top rim:

$$A_3 = \pi(21)^2 - \pi(19)^2 = \pi(441 - 361) = 80\pi$$

5. Total surface area A total

Total surface area:

$$A_{\text{total}} = 882\pi + 722\pi + 80\pi = 1684\pi$$

6. Note

Three separate surfaces! Missing the flat annular ring is the most common error on this type of question.

Final answer

$$\text{Total surface area} = 1684\pi \text{ cm}^2$$

CLASSIFIED PROBLEM

Paper: 4WM2H

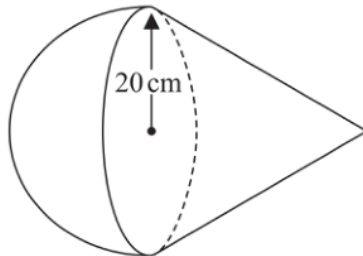
Session: Specimen

Q#: 22

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Specimen 4WM2H

22 A solid is made from a cone and a hemisphere.Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

 $k = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM2H | Question 22

Volume & Surface Area

1. Calculate surface area

The curved surface area of a cone is

$$\pi r l$$

2. Calculate areaHere $r = 20$ and $\pi r l = 580\pi$, so

$$20l = 580$$

$$l = 29$$

3. Use Pythagoras to find the

Use Pythagoras to find the height of the cone:

$$h^2 = 29^2 - 20^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

4. Volume of the cone

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

5. Volume of the hemisphere

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

6. Total volume

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

7. Calculate areaSince the volume is $k\pi$,

$$k = \frac{24400}{3}$$

Final answer

$$k = \frac{24400}{3}.$$

EA

Area & Volume of Similar Shapes

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

Q#: 25

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: May 2025 4WM2H

25 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 25 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 25

Area & Volume of Similar Shapes

1. Linear scale factor from solid

Linear scale factor from solid A to solid B :

$$\frac{18.6}{31} = 0.6$$

2. Calculate volume

So the volume scale factor is

$$0.6^3 = 0.216$$

3. Let the volume of be

Let the volume of A be V . Then

$$V - 0.216V = 735$$

$$0.784V = 735$$

$$V = 937.5$$

Final answer937.5 cm³.

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Nov2025

Q#: 23

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Source: Nov 2025 4WM2H

23 Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape **Q**..... cm³

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 23

Area & Volume of Similar Shapes

1. Calculate area

$$ASF = \frac{\text{Area of Q}}{\text{Area of P}} = \frac{450}{200} = \frac{9}{4} = 2.25$$

2. Simplify surd

$$LSF = \sqrt{ASF} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$$

3. Calculate volume

$$VSF = (LSF)^3 = \left(\frac{3}{2}\right)^3 = \frac{27}{8} = 3.375$$

4. Calculate volume

$$\begin{aligned} \text{Volume of Q} &= \text{Volume of P} \times VSF = 672 \times \frac{27}{8} \\ &= 84 \times 27 = 2268 \end{aligned}$$

Final answer2268 cm³

Volume & Surface Area

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2020

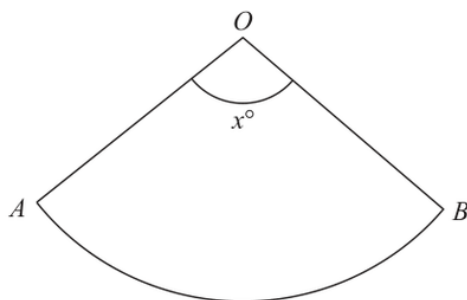
Q#: 26

Marks: 6

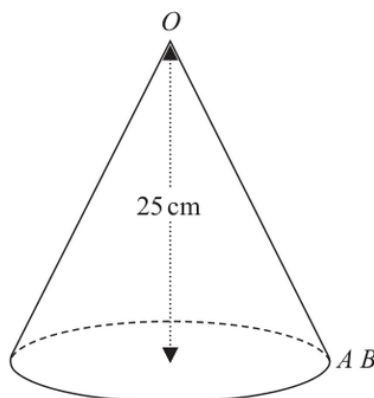
TOPIC: Volume & Surface Area

Source: Jan 2020 P2H

- 26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

Diagram **NOT**
accurately drawn

The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of x .
Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

(Total for Question 26 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 26

Volume & Surface Area

1. Let the radius of theLet the radius of the base of the cone be r , and let the slant height be l .**2. The volume of the cone**

The volume of the cone is

$$\frac{1}{3}\pi r^2 h = 1600$$

3. Calculate area

The height is 25 cm, so

$$\frac{1}{3}\pi r^2 (25) = 1600$$

$$r^2 = \frac{4800}{25\pi}$$

$$r = 7.8176\dots$$

4. The slant height is found

The slant height is found using Pythagoras:

$$l^2 = r^2 + 25^2$$

$$l = 26.1938\dots$$

5. Calculate area

When the sector is joined to form the cone, the arc length of the sector becomes the circumference of the cone base.

$$\frac{x}{360} \times 2\pi l = 2\pi r$$

6. CancelCancel 2π :

$$\frac{x}{360} l = r$$

$$x = \frac{360r}{l}$$

$$x = \frac{360(7.8176\dots)}{26.1938\dots} = 107.4433\dots$$

Final answer**107°**

CLASSIFIED PROBLEM

Paper: P1H

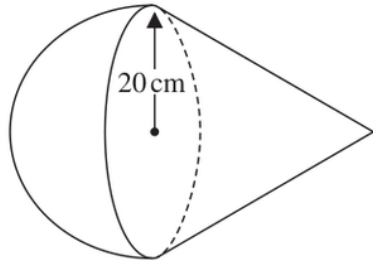
Session: Jun 2022

Q#: 22

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Jun 2022 P1H

22 A solid is made from a cone and a hemisphere.Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
 The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi\text{cm}^2$

The volume of the solid is $k\pi\text{cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 22

Volume & Surface Area

1. The cone has radius cm

The cone has radius 20 cm.

2. Calculate surface area

The curved surface area of the cone is

$$\pi r l$$

3. Calculate value

So

$$\pi(20)l = 580\pi$$

$$l = 29$$

4. Use Pythagoras to find the

Use Pythagoras to find the height of the cone:

$$h^2 + 20^2 = 29^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

5. Volume of the cone

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

6. Volume of the hemisphere

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

7. Total volume

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

8. Calculate volumeThe volume is $k\pi$, so

$$k = \frac{24400}{3}$$

Final answer

$\frac{24400}{3}$

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2023

Q#: 24

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Jun 2023 P2HR

- 24** The surface area of sphere **A** is nine times the surface area of sphere **B**
The difference between the volume of sphere **A** and the volume of sphere **B** is $117\pi \text{ cm}^3$

Find the radius of the smaller sphere.

Show your working clearly.

..... cm

(Total for Question 24 is 5 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 24
Volume & Surface Area

1. Let the radius of the

Let the radius of the smaller sphere be r .

2. Let the radius of the

Let the radius of the larger sphere be R .

3. Surface area of a sphere

Surface area of a sphere is

$$4\pi r^2$$

4. The surface area of sphere

The surface area of sphere A is nine times the surface area of sphere B , so

$$4\pi R^2 = 9(4\pi r^2)$$

$$R^2 = 9r^2$$

$$R = 3r$$

5. The difference between the volumes

The difference between the volumes is 117π :

$$\frac{4}{3}\pi(3r)^3 - \frac{4}{3}\pi r^3 = 117\pi$$

$$\frac{4}{3}\pi(27r^3 - r^3) = 117\pi$$

$$\frac{104}{3}\pi r^3 = 117\pi$$

6. Cancel

Cancel π :

$$\frac{104}{3}r^3 = 117$$

$$r^3 = \frac{117 \times 3}{104}$$

$$r^3 = \frac{27}{8}$$

$$r = \frac{3}{2}$$

Final answer

1.5 cm

EA

CLASSIFIED PROBLEM

Paper: P2H

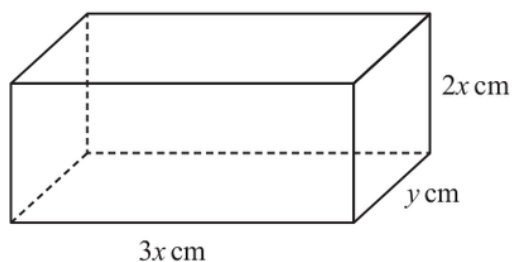
Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: **Volume & Surface Area**

Source: Jun 2025 P2H

21 The diagram shows a cuboid.Diagram **NOT**
accurately drawnThe cuboid measures $3x$ cm by $2x$ cm by y cmThe volume of the cuboid is 1014 cm^3 The total surface area of the cuboid is $A\text{ cm}^2$ Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 21

Volume & Surface Area

1. The cuboid has dimensions

The cuboid has dimensions

$$3x, \quad 2x, \quad y$$

2. Calculate volumeIts volume is 1014 cm^3 , so

$$(3x)(2x)(y) = 1014$$

$$6x^2y = 1014$$

$$y = \frac{1014}{6x^2}$$

$$y = \frac{169}{x^2}$$

3. The total surface area is

The total surface area is

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$A = 2(6x^2 + 5xy)$$

$$A = 12x^2 + 10xy$$

4. SubstituteSubstitute $y = \frac{169}{x^2}$:

$$A = 12x^2 + 10x \left(\frac{169}{x^2} \right)$$

$$A = 12x^2 + \frac{1690}{x}$$

Final answer

$$A = 12x^2 + \frac{1690}{x}$$

CLASSIFIED PROBLEM

Paper: P2HR

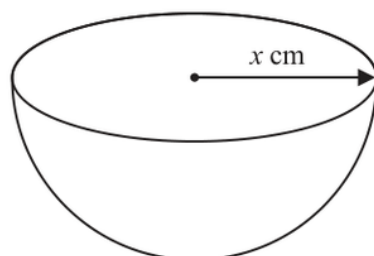
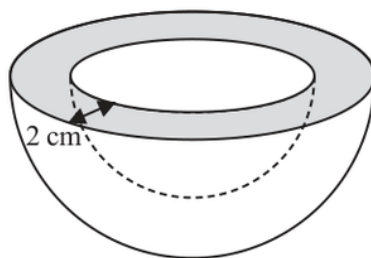
Session: Jun 2025

Q#: 26

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Jun 2025 P2HR

26 The diagram shows a solid hemisphere, H Diagram **NOT**
accurately drawnThe radius of H is x cmThe volume of H is 6174π cm³A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cmDiagram **NOT**
accurately drawnWork out the **total** surface area of the bowl.Give your answer in terms of π cm²**(Total for Question 26 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 26

Volume & Surface Area

1. The volume of a hemisphere

The volume of a hemisphere is

$$\frac{2}{3}\pi r^3$$

2. Evaluate fractionFor the original hemisphere H ,

$$\frac{2}{3}\pi x^3 = 6174\pi$$

3. CancelCancel π :

$$\frac{2}{3}x^3 = 6174$$

$$x^3 = 9261$$

$$x = 21$$

4. The outer radius of the

The outer radius of the bowl is 21 cm.

5. Calculate value

The thickness is 2 cm, so the inner radius is

$$21 - 2 = 19$$

6. Total surface area of the

Total surface area of the bowl is:

7. outer curved hemisphere inner curved

- outer curved hemisphere
- inner curved hemisphere
- flat annular rim

8. Outer curved surface area

Outer curved surface area:

$$2\pi(21)^2 = 882\pi$$

9. Inner curved surface area

Inner curved surface area:

$$2\pi(19)^2 = 722\pi$$

10. Flat rim area

Flat rim area:

$$\pi(21^2 - 19^2) = 80\pi$$

11. Total

Total:

$$882\pi + 722\pi + 80\pi = 1684\pi$$

Final answer

$$1684\pi \text{ cm}^2$$

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2024

Q#: 23

Marks: 5

TOPIC: Volume & Surface Area

Source: May 2024 P1H

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

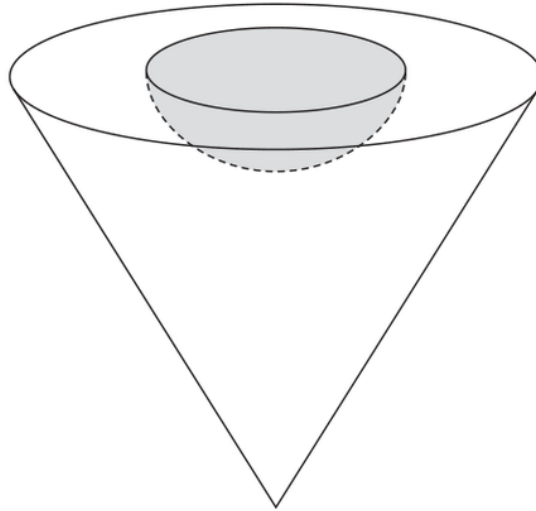


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 23

Volume & Surface Area

1. Define cone dimensionsThe cone has radius $5x$ and height $6x$.**2. Volume of the cone**

Volume of the cone:

$$\begin{aligned}\frac{1}{3}\pi(5x)^2(6x) \\ = 50\pi x^3\end{aligned}$$

3. The removed hemisphere has radiusThe removed hemisphere has radius $2x$.**4. Volume of the hemisphere**

Volume of the hemisphere:

$$\begin{aligned}\frac{2}{3}\pi(2x)^3 \\ = \frac{16}{3}\pi x^3\end{aligned}$$

5. The remaining solid has volumeThe remaining solid has volume 6948π , so

$$\begin{aligned}50\pi x^3 - \frac{16}{3}\pi x^3 &= 6948\pi \\ \frac{134}{3}\pi x^3 &= 6948\pi\end{aligned}$$

6. CancelCancel π :

$$\begin{aligned}\frac{134}{3}x^3 &= 6948 \\ x^3 &= \frac{10422}{67} \\ x &= 5.3781\dots\end{aligned}$$

7. The solid hemisphere removed hasThe solid hemisphere removed has radius $2x$.**8. Total surface area of a**

Total surface area of a solid hemisphere is

$$3\pi r^2$$

9. Calculate value

So

$$3\pi(2x)^2 = 12\pi x^2$$

$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$

Final answer

$$1090 \text{ cm}^2$$

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2021

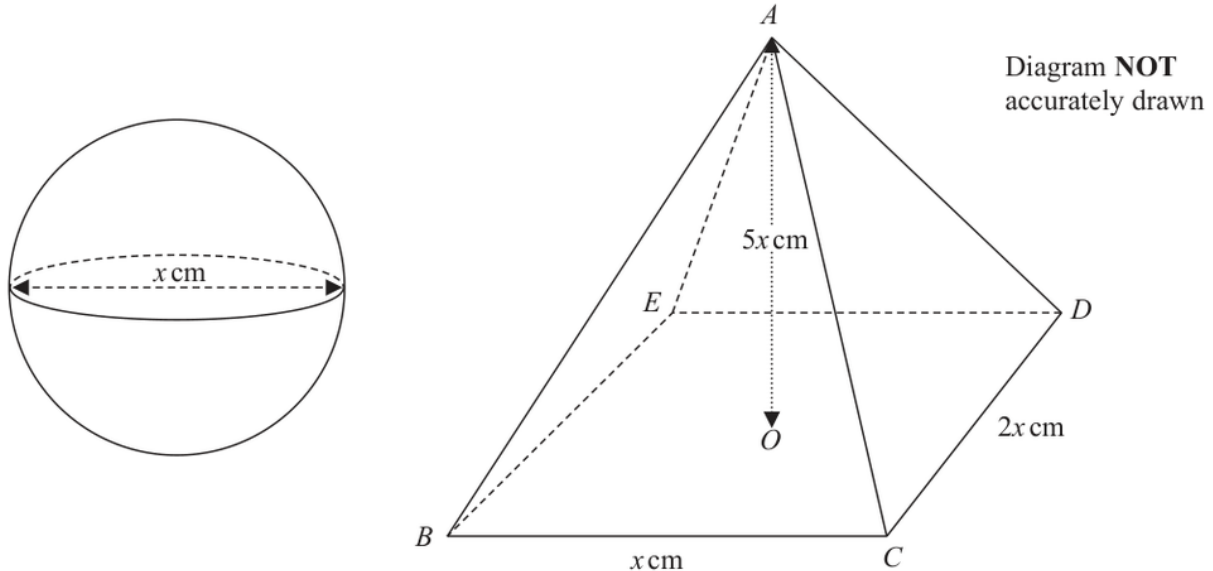
Q#: 22

Marks: 6

TOPIC: Volume & Surface Area

Source: Nov 2021 P2H

- 22** The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is 288π cm³

Calculate the total surface area of the pyramid.

Give your answer correct to the nearest cm²

..... cm²

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P2H | Question 22

Volume & Surface Area

1. Use trigonometryThe sphere has diameter x , so its radius is

$$\frac{x}{2}$$

2. The volume of the sphereThe volume of the sphere is $288\pi \text{ cm}^3$:

$$\frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = 288\pi$$

$$\frac{\pi x^3}{6} = 288\pi$$

$$x^3 = 1728$$

$$x = 12$$

3. The rectangular base of theThe rectangular base of the pyramid has dimensions x and $2x$, so its area is

$$x(2x) = 2x^2$$

4. Use trigonometryFor the two faces with base x , the slant height is

$$\sqrt{(5x)^2 + x^2} = x\sqrt{26}$$

5. Their total area is

Their total area is

$$2\left(\frac{1}{2}x \cdot x\sqrt{26}\right) = x^2\sqrt{26}$$

6. Use trigonometryFor the two faces with base $2x$, the slant height is

$$\sqrt{(5x)^2 + \left(\frac{x}{2}\right)^2} = \frac{x\sqrt{101}}{2}$$

7. Their total area is

Their total area is

$$2\left(\frac{1}{2}(2x) \cdot \frac{x\sqrt{101}}{2}\right) = x^2\sqrt{101}$$

8. Total surface area

Total surface area:

$$2x^2 + x^2\sqrt{26} + x^2\sqrt{101}$$

9. SubstituteSubstitute $x = 12$:

$$144(2 + \sqrt{26} + \sqrt{101}) = 2469.440 \dots$$

10. Correct to the nearestCorrect to the nearest cm^2 :

$$2469$$

Final answer

$$2469 \text{ cm}^2$$

EA

CLASSIFIED PROBLEM

Paper: P1H

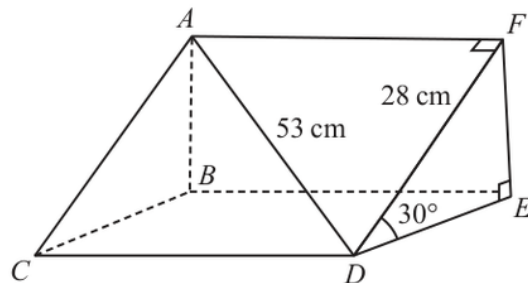
Session: Nov 2023

Q#: 21

Marks: 5

TOPIC: Volume & Surface Area

Source: Nov 2023 P1H

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 21

Volume & Surface Area

1. The triangular cross section isThe triangular cross-section is triangle DEF .**2. It is right angled at**It is right-angled at E , with

$$DF = 28$$

3. Use trigonometry

and

$$\angle FDE = 30^\circ$$

4. Use trigonometry

So

$$FE = 28 \sin 30^\circ = 14$$

5. Use trigonometry

and

$$DE = 28 \cos 30^\circ = 14\sqrt{3}$$

6. Area of triangleArea of triangle DEF :

$$\frac{1}{2}(14)(14\sqrt{3}) = 98\sqrt{3}$$

7. The length of the prism

The length of the prism is

$$AD = 53$$

8. Volume of the prism

Volume of the prism:

$$98\sqrt{3} \times 53$$

$$= 5194\sqrt{3}$$

$$= 8996.2719 \dots$$

Final answer 8996 cm^3

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2024

Q#: 22

Marks: 3

TOPIC: **Volume & Surface Area**

Source: Nov 2024 P1H

- 22** The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

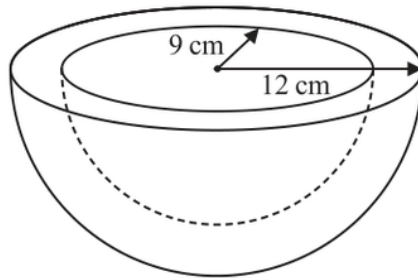


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm

The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm³

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 22

Volume & Surface Area

1. The bowl is a hemispherical

The bowl is a hemispherical shell.

2. The volume of the metal

The volume of the metal is:

$$\text{outer hemisphere volume} - \text{inner hemisphere volume}$$

3. Volume of a hemisphere

Volume of a hemisphere:

$$\frac{2}{3}\pi r^3$$

4. Calculate volume

So

$$\text{metal volume} = \frac{2}{3}\pi(12)^3 - \frac{2}{3}\pi(9)^3$$

$$= \frac{2}{3}\pi(12^3 - 9^3)$$

$$= \frac{2}{3}\pi(1728 - 729)$$

$$= 666\pi$$

$$= 2092.3007\dots$$

Final answer

$$2092 \text{ cm}^3$$

Area & Volume of Similar Shapes

EA

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jan 2020

Q#: 20

Marks: 5

TOPIC: Area & Volume of Similar Shapes

Source: Jan 2020 P1HR

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

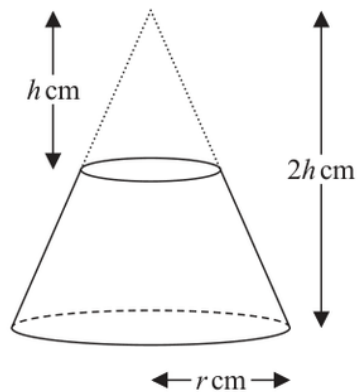
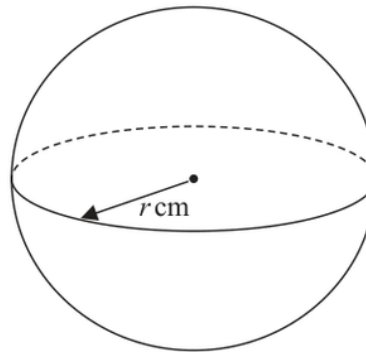


Diagram **NOT**
accurately drawn



The height of the small cone is h cm.
The height of the large cone is $2h$ cm.
The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h .

Give your expression in its simplest form.

$r = \dots\dots\dots$

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 20

Area & Volume of Similar Shapes

1. The two cones are similar

The two cones are similar.

2. The large cone has heightThe large cone has height $2h$ and base radius r .**3. The small cone has height**The small cone has height h , so its linear scale factor compared with the large cone is

$$\frac{h}{2h} = \frac{1}{2}$$

4. Define cone dimensions

So the small cone has radius

$$\frac{r}{2}$$

5. Volume of the large cone

Volume of the large cone:

$$\frac{1}{3}\pi r^2(2h) = \frac{2}{3}\pi r^2h$$

6. Volume of the small cone

Volume of the small cone:

$$\frac{1}{3}\pi \left(\frac{r}{2}\right)^2 h = \frac{1}{12}\pi r^2h$$

7. Calculate volume

So the volume of the frustum is

$$\frac{2}{3}\pi r^2h - \frac{1}{12}\pi r^2h = \frac{7}{12}\pi r^2h$$

8. Calculate volumeThe sphere has radius r , so its volume is

$$\frac{4}{3}\pi r^3$$

9. The two volumes are equal

The two volumes are equal:

$$\frac{7}{12}\pi r^2h = \frac{4}{3}\pi r^3$$

10. Divide by

Divide by πr^2 :

$$\frac{7}{12}h = \frac{4}{3}r$$

$$7h = 16r$$

$$r = \frac{7h}{16}$$

Final answer

$$r = \frac{7h}{16}$$

EA

CLASSIFIED PROBLEM

Paper: P2HR

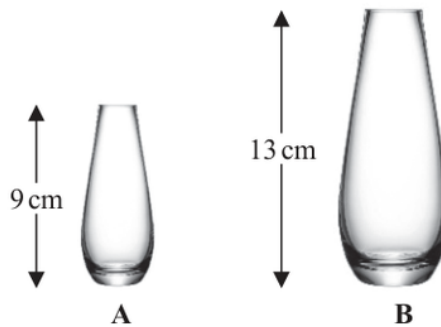
Session: Jan 2020

Q#: 20

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: Jan 2020 P2HR

20 The diagram shows two similar vases, **A** and **B**.Diagram **NOT**
accurately drawnThe height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**...... cm²

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 20

Area & Volume of Similar Shapes

1. The two vases are similar

The two vases are similar.

2. Their heights are in the

Their heights are in the ratio

$$9 : 13$$

3. Split the ratio

So their surface areas are in the ratio

$$9^2 : 13^2 = 81 : 169$$

4. The total surface area isThe total surface area is 1800 cm^2 , so there are

$$81 + 169 = 250$$

5. parts altogether

parts altogether.

6. Surface area of vase A

Surface area of vase A:

$$1800 \times \frac{81}{250} = 583.2$$

Final answer

$$583.2 \text{ cm}^2$$

CLASSIFIED PROBLEM

Paper: P1H

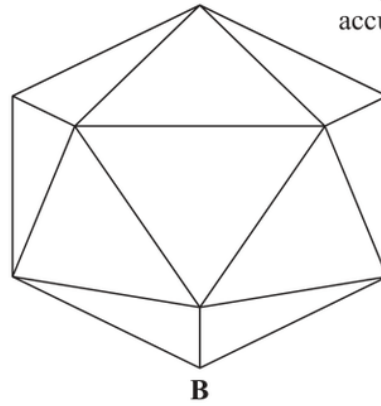
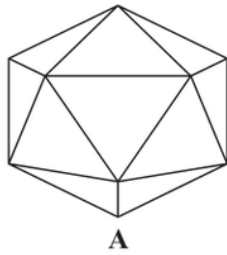
Session: Jan 2021

Q#: 20

Marks: 3

TOPIC: Area & Volume of Similar Shapes

Source: Jan 2021 P1H

20 **A** and **B** are two similar solids.Diagram **NOT**
accurately drawn**A** has a volume of 1836 cm^3 **B** has a volume of 4352 cm^3 **B** has a total surface area of 1120 cm^2 Work out the total surface area of **A**...... cm^2

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 20

Area & Volume of Similar Shapes

1. The volume ratio of solid

The volume ratio of solid A to solid B is

$$1836 : 4352$$

2. Simplify

Simplify:

$$\frac{1836}{4352} = \frac{27}{64}$$

3. Split the ratio

So the linear ratio is

$$\sqrt[3]{27} : \sqrt[3]{64} = 3 : 4$$

4. Surface area ratio

Surface area ratio:

$$3^2 : 4^2 = 9 : 16$$

5. Solid B has surface area

Solid B has surface area 1120 cm^2 , so solid A has surface area

$$1120 \times \frac{9}{16} = 630$$

Final answer

$$630 \text{ cm}^2$$

CLASSIFIED PROBLEM

Paper: P1HR

Session: Jun 2022

Q#: 20

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: Jun 2022 P1HR

20 The diagram shows two similar metal statues.**A****B**Diagram **NOT**
accurately drawnThe volume of statue **B** is 20% less than the volume of statue **A**The surface area of statue **B** is $k\%$ less than the surface area of statue **A**Work out the value of k

Give your answer correct to 3 significant figures.

 $k = \dots\dots\dots$ **(Total for Question 20 is 4 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 20

Area & Volume of Similar Shapes

1. The volume of statue is

The volume of statue B is 20% less than statue A , so the volume scale factor is 0.8.

2. Calculate area

For similar solids, the linear scale factor is the cube root of the volume scale factor:

$$\text{linear scale factor} = \sqrt[3]{0.8}$$

3. Surface area scale factor is

Surface area scale factor is the square of the linear scale factor:

$$\text{surface area scale factor} = (\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773\dots$$

4. Calculate area

So statue B has 86.1773...% of the surface area of statue A .

$$k = 100 - 86.1773\dots = 13.8226\dots$$

Final answer

$k = 13.8$ to 3 significant figures.

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 24

Marks: 6

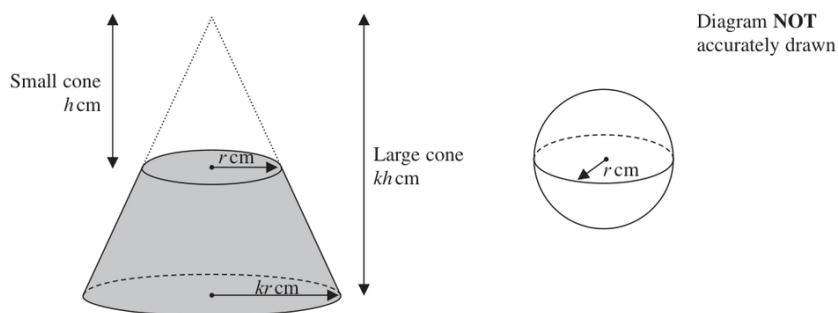
TOPIC: Area & Volume of Similar Shapes

Source: Jun 2022 P2HR

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.



The height of the small cone is h cm and the radius of the base of the small cone is r cm.
 The height of the large cone is kh cm and the radius of the base of the large cone is kr cm.
 The radius of the sphere is r cm.

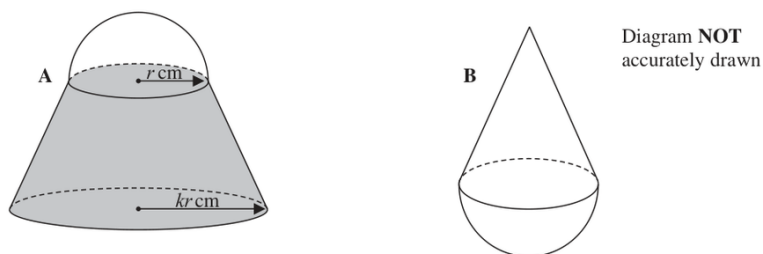
The sphere is divided into two hemispheres, each of radius r cm.

Solid A is formed by joining one of the hemispheres to the frustum.

The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid B is formed by joining the other hemisphere to the small cone that was removed from the large cone.

The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.



The volume of solid A is 6 times the volume of solid B.

Given that $k > \sqrt[3]{7}$

find an expression for h in terms of k and r

$h = \dots\dots\dots$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 24

Area & Volume of Similar Shapes

1. Small cone volume

Small cone volume:

$$V_s = \frac{1}{3}\pi r^2 h$$

2. The larger similar cone has

The larger similar cone has radius kr and height kh , so its volume is

$$V_L = \frac{1}{3}\pi(kr)^2(kh) = \frac{1}{3}\pi k^3 r^2 h$$

3. Calculate volume

So the frustum volume is

$$V_L - V_s = \frac{1}{3}\pi r^2 h(k^3 - 1)$$

4. The volume of one hemisphere

The volume of one hemisphere of radius r is

$$\frac{2}{3}\pi r^3$$

5. Solid A is the frustum

Solid A is the frustum plus one hemisphere:

$$V_A = \frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3$$

6. Solid B is the small

Solid B is the small cone plus one hemisphere:

$$V_B = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

7. Given

Given $V_A = 6V_B$:

$$\frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3 = 6\left(\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3\right)$$

8. Divide by

Divide by $\frac{1}{3}\pi r^2$:

$$h(k^3 - 1) + 2r = 6h + 12r$$

$$h(k^3 - 7) = 10r$$

$$h = \frac{10r}{k^3 - 7}$$

Final answer

$$h = \frac{10r}{k^3 - 7}$$

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2024

Q#: 24

Marks: 5

TOPIC: Area & Volume of Similar Shapes

Source: Jun 2024 P2HR

- 24 The diagram shows a solid, **S**, made from a cone and a hemisphere.

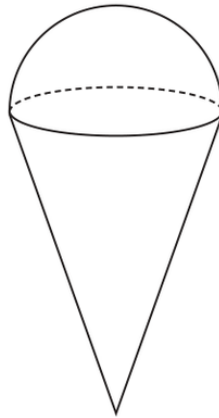


Diagram **NOT**
accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is $4 \times$ the radius of the hemisphere.

A separate sphere has radius kx cm.

The volume of this sphere is $12.5 \times$ the volume of **S**

- (a) Work out the value of k

$$k = \dots\dots\dots (4)$$

A solid, **T**, is similar to solid **S**

The volume of **T** is $512 \times$ the volume of **S**

The total surface area of **T** is $d \times$ the total surface area of **S**

- (b) Find the value of d

$$d = \dots\dots\dots (1)$$

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 24

Area & Volume of Similar Shapes

1. The hemisphere has radiusThe hemisphere has radius x , so its height is x .**2. The total height of solid**The total height of solid S is $4x$, so the cone height is

$$4x - x = 3x$$

3. Volume of the cone

Volume of the cone:

$$\frac{1}{3}\pi x^2(3x) = \pi x^3$$

4. Volume of the hemisphere

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

5. Calculate volume

So the volume of solid S is

$$\pi x^3 + \frac{2}{3}\pi x^3 = \frac{5}{3}\pi x^3$$

6. Calculate volumeThe sphere has radius kx , so its volume is

$$\frac{4}{3}\pi(kx)^3 = \frac{4}{3}\pi k^3 x^3$$

7. This is times the volume

This is 12.5 times the volume of S:

$$\frac{4}{3}\pi k^3 x^3 = 12.5 \left(\frac{5}{3}\pi x^3 \right)$$

8. CancelCancel $\frac{1}{3}\pi x^3$:

$$4k^3 = 62.5$$

$$k^3 = 15.625 = \frac{125}{8}$$

$$k = 2.5$$

9. Calculate volume

For solid T, the volume scale factor is 512.

$$512 = 8^3$$

10. Calculate area

So the linear scale factor is 8, and the surface area scale factor is

$$8^2 = 64$$

Final answer

$$k = 2.5, d = 64$$

EA

CLASSIFIED PROBLEM

Paper: P1HR

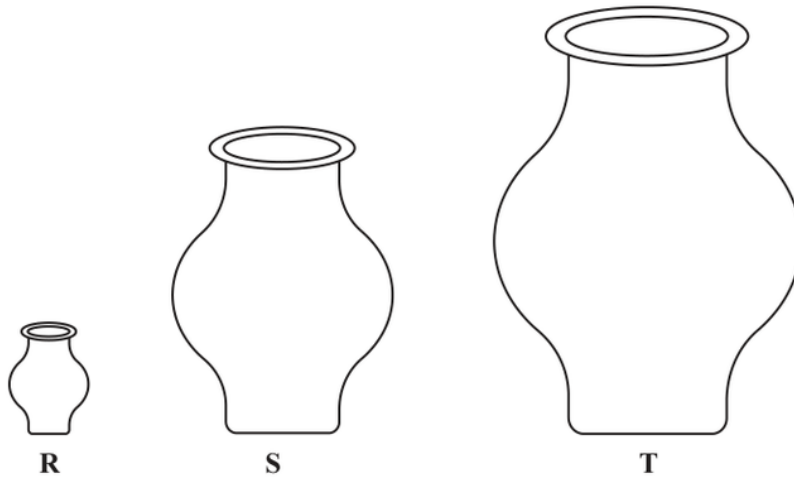
Session: Jun 2025

Q#: 26

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: Jun 2025 P1HR

26 **R, S and T** are three similar vases.Diagram **NOT**
accurately drawnThe volume of vase **S** is 72.8% more than the volume of vase **R**The height of vase **R** is h cmThe height of vase **T** is $6h$ cmThe surface area of vase **S** is A cm²The surface area of vase **T** is kA cm²Work out the value of k $k = \dots\dots\dots$ **(Total for Question 26 is 4 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 26

Area & Volume of Similar Shapes

1. The volume of vase isThe volume of vase S is 72.8% more than vase R , so

$$\frac{V_S}{V_R} = 1.728$$

2. Calculate area

For similar solids, volume scale factor is the cube of the linear scale factor:

linear scale factor from R to $S = \sqrt[3]{1.728} = 1.2$ **3. The height of vase is**The height of vase T is $6h$, so the linear scale factor from R to T is 6.**4. Calculate area**Therefore the linear scale factor from S to T is

$$\frac{6}{1.2} = 5$$

5. Surface area scale factor is

Surface area scale factor is the square of the linear scale factor:

$$k = 5^2 = 25$$

Final answer

$$k = 25$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2025

Q#: 26

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Source: Jun 2025 P2H

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 26

Area & Volume of Similar Shapes

1. The heights of solids A

The heights of solids A and B are 31 cm and 18.6 cm.

$$31 : 18.6 = 5 : 3$$

2. Split the ratio

So the volumes are in the ratio

$$5^3 : 3^3 = 125 : 27$$

3. The difference in volume is

The difference in volume is

$$125 - 27 = 98$$

4. parts

parts.

5. Evaluate fractionThis difference is 735 cm^3 , so one part is

$$\frac{735}{98} = 7.5$$

6. Volume of solid A

Volume of solid A:

$$125 \times 7.5 = 937.5$$

Final answer

$$937.5 \text{ cm}^3$$

CLASSIFIED PROBLEM

Paper: P2H

Session: May 2021

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Source: May 2021 P2H

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3 **(Total for Question 20 is 3 marks)**

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 20

Area & Volume of Similar Shapes

1. Split the ratio

The surface area ratio of the small block to the large block is

$$25 : 36$$

2. Split the ratio

So the linear ratio is

$$\sqrt{25} : \sqrt{36} = 5 : 6$$

3. Split the ratio

Therefore the volume ratio is

$$5^3 : 6^3 = 125 : 216$$

4. The small block has volumeThe small block has volume 300 cm^3 .**5. Large block volume**

Large block volume:

$$300 \times \frac{216}{125} = 518.4$$

Final answer518.4 cm³

CLASSIFIED PROBLEM

Paper: P1H

Session: May 2023

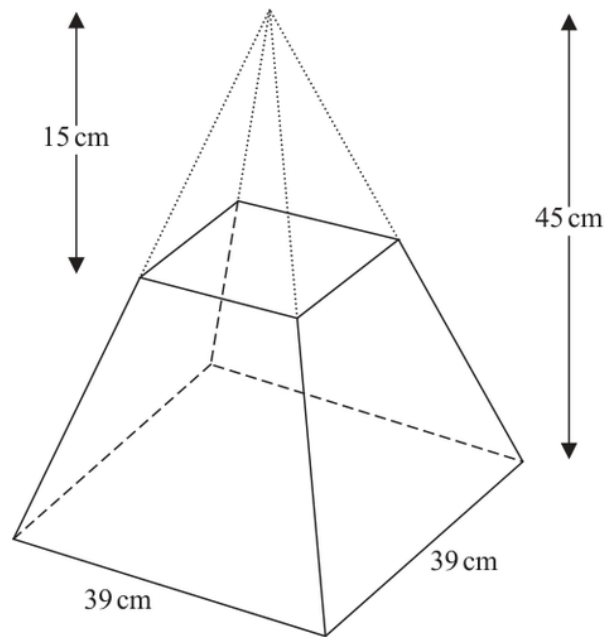
Q#: 23

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Source: May 2023 P1H

- 23** A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

..... cm²

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 23

Area & Volume of Similar Shapes

1. The small and large square

The small and large square-based pyramids are similar.

2. Their heights are

Their heights are

$$15 : 45 = 1 : 3$$

3. Square of the frustum

So the side length of the top square of the frustum is

$$\frac{39}{3} = 13 \text{ cm}$$

4. The vertical height of the

The vertical height of the frustum is

$$45 - 15 = 30 \text{ cm}$$

5. Evaluate fraction

For one trapezium face, the difference between the half-side lengths is

$$\frac{39 - 13}{2} = 13 \text{ cm}$$

6. Simplify surd

So the slant height of the trapezium face is

$$\sqrt{30^2 + 13^2} = \sqrt{1069}$$

7. Area of one trapezium face

Area of one trapezium face:

$$\frac{1}{2}(39 + 13)\sqrt{1069} = 26\sqrt{1069}$$

8. Area of four trapezium faces

Area of four trapezium faces:

$$4(26\sqrt{1069}) = 104\sqrt{1069}$$

9. Add the top and bottom

Add the top and bottom square faces:

$$13^2 + 39^2 = 169 + 1521 = 1690$$

10. Total surface area

Total surface area:

$$1690 + 104\sqrt{1069} = 5090.338 \dots$$

11. Correct to the nearest whole

Correct to the nearest whole number:

5090

Final answer

5090 cm²

EA

CLASSIFIED PROBLEM

Paper: P1H

Session: MayNov 2020

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Source: May-Nov 2020 P1H

20 **R** and **S** are two similar solid shapes.Shape **R** has surface area 108 cm^2 and volume 135 cm^3 Shape **S** has surface area 300 cm^2 Work out the volume of shape **S**...... cm^3

(Total for Question 20 is 3 marks)

EA

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 20

Area & Volume of Similar Shapes

1. Shape R has surface areaShape R has surface area 108 cm^2 and volume 135 cm^3 .**2. Shape S has surface area**Shape S has surface area 300 cm^2 .**3. Surface area ratio S : R**

Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

4. Split the ratio

So the linear ratio S:R is

$$5 : 3$$

5. Split the ratio

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

6. Volume of S

Volume of S:

$$135 \times \frac{125}{27} = 625$$

Final answer

$$625 \text{ cm}^3$$

CLASSIFIED PROBLEM

TOPIC: Area & Volume of Similar Shapes

Source: Nov 2023 P1H

Paper: P1H

Session: Nov 2023

Q#: 23

Marks: 5

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

..... cm

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 23

Area & Volume of Similar Shapes

1. The area ratio of solid

The area ratio of solid A to solid B is

$$7776 : 486 = 16 : 1$$

2. Split the ratio

So the linear ratio A:B is

$$\sqrt{16} : 1 = 4 : 1$$

3. Split the ratio

Therefore the volume ratio A:B is

$$4^3 : 1^3 = 64 : 1$$

4. Calculate volumeThe volumes are 8^x and 2^{x+4} , so

$$\frac{8^x}{2^{x+4}} = 64$$

5. Use index lawsWrite 8^x as a power of 2:

$$8^x = (2^3)^x = 2^{3x}$$

6. Evaluate fraction

So

$$\frac{2^{3x}}{2^{x+4}} = 2^6$$

$$2^{3x-(x+4)} = 2^6$$

$$2x - 4 = 6$$

$$x = 5$$

7. The height of solid A

The height of solid A is

$$3^x = 3^5 = 243$$

8. Split the ratio

Since the linear ratio A:B is 4 : 1, the height of solid B is

$$\frac{243}{4} = 60.75$$

Final answer

60.75 cm

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2023

Q#: 23

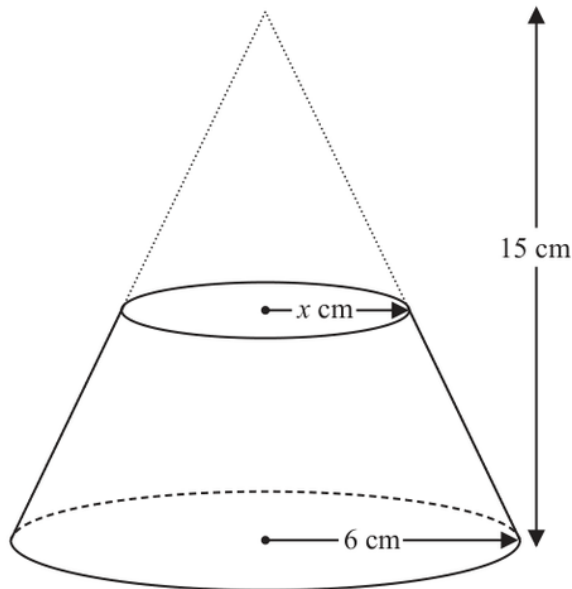
Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Source: Nov 2023 P2H

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

Diagram **NOT**
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$ work out the value of x

Show clear algebraic working.

 $x = \dots\dots\dots$ **(Total for Question 23 is 5 marks)**

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 23

Area & Volume of Similar Shapes

1. The large cone has height

The large cone has height 15 cm and radius 6 cm.

2. The small cone has radiusThe small cone has radius x . Since the cones are similar,

$$\frac{\text{radius}}{\text{height}} = \frac{6}{15} = \frac{2}{5}$$

3. Evaluate fraction

So for the small cone,

$$\frac{x}{h} = \frac{2}{5}$$

$$h = \frac{5}{2}x$$

4. Volume of the large cone

Volume of the large cone:

$$\frac{1}{3}\pi(6)^2(15) = 180\pi$$

5. Volume of the small cone

Volume of the small cone:

$$\frac{1}{3}\pi x^2 \left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$$

6. The frustum volume is

The frustum volume is

$$\frac{4212}{25}\pi$$

7. Evaluate fraction

So

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi$$

8. CancelCancel π :

$$180 - \frac{5}{6}x^3 = \frac{4212}{25}$$

$$\frac{5}{6}x^3 = 180 - \frac{4212}{25}$$

$$\frac{5}{6}x^3 = \frac{288}{25}$$

$$x^3 = \frac{1728}{125}$$

$$x = \frac{12}{5} = 2.4$$

Final answer

2.4 cm

EA

CLASSIFIED PROBLEM

TOPIC: Area & Volume of Similar Shapes

Source: Nov 2025 P2H

Paper: P2H

Session: Nov 2025

Q#: 23

Marks: 3

23 Shape P is similar to shape Q

The table shows some information about shape P and shape Q

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape Q

..... cm³

(Total for Question 23 is 3 marks)

WORKED SOLUTION*Dr. Eslam Ahmed***Nov 2025 P2H | Question 23**

Area & Volume of Similar Shapes

1. Surface area ratio Q P

Surface area ratio Q:P:

$$450 : 200 = 9 : 4$$

2. Split the ratio

So the linear ratio Q:P is

$$3 : 2$$

3. Split the ratio

Therefore the volume ratio Q:P is

$$3^3 : 2^3 = 27 : 8$$

4. Shape P has volumeShape P has volume 672 cm^3 .**5. Volume of Q**

Volume of Q:

$$672 \times \frac{27}{8} = 2268$$

Final answer**2268 cm³**

Vectors

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

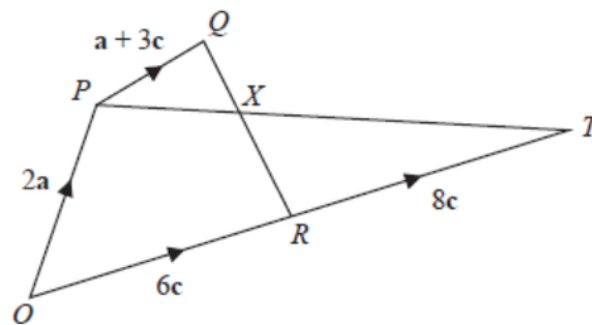
Q#: 26

Marks: 5

TOPIC: Vectors

Source: May 2025 4WM2H

26

Diagram NOT
accurately drawn

The diagram shows a quadrilateral $OPQR$ in which

$$\vec{OP} = 2\mathbf{a} \quad \vec{OR} = 6\mathbf{c} \quad \vec{PQ} = \mathbf{a} + 3\mathbf{c}$$

(a) Find $\vec{QR}$ in terms of $\mathbf{a}$ and $\mathbf{c}$

$$\vec{QR} = \dots\dots\dots (1)$$

The point T is such that $\vec{RT} = 8\mathbf{c}$
 The point X lies on QR such that PXT is a straight line.

(b) Using a vector method, find the ratio $QX : XR$
 Give your answer in the form $m : n$ where m and n are integers.
 Show your working clearly.

P81641A

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$$QX : XR = \dots\dots\dots (4)$$

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 26

Vectors

1. (a)

(a)

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$$

$$\overrightarrow{OQ} = 2a + (a + 3c) = 3a + 3c$$

$$\overrightarrow{QR} = 6c - (3a + 3c) = 3c - 3a$$

2. (b) Let divide so that

(b) Let X divide QR so that

$$\overrightarrow{QX} = \lambda \overrightarrow{QR}$$

3. Calculate value

Then

$$\overrightarrow{OX} = 3a + 3c + \lambda(3c - 3a)$$

4. Calculate value

Also X lies on PT , where $\overrightarrow{OT} = 14c$, so

$$\overrightarrow{OX} = 2a + \mu(14c - 2a)$$

5. Equating coefficients gives . Therefore

Equating coefficients gives $\lambda = \frac{5}{9}$. Therefore

$$QX : XR = \frac{5}{9} : \frac{4}{9} = 5 : 4$$

Final answer

$$\overrightarrow{QR} = 3c - 3a; QX : XR = 5 : 4.$$

CLASSIFIED PROBLEM

Paper: 4WM2H

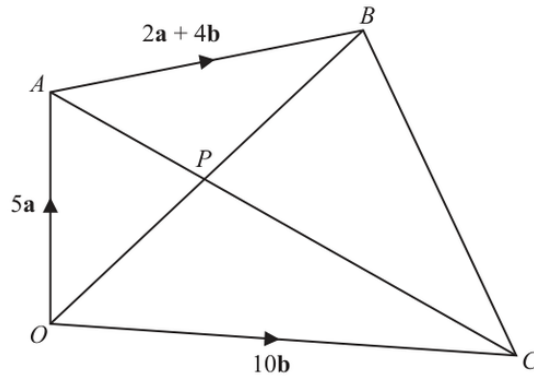
Session: Nov2025

Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Nov 2025 4WM2H

24 The diagram shows a quadrilateral $OABC$ Diagram **NOT**
accurately drawn

$$\vec{OA} = 5\mathbf{a} \quad \vec{OC} = 10\mathbf{b} \quad \vec{AB} = 2\mathbf{a} + 4\mathbf{b}$$

(a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AC} = \dots\dots\dots (1)$$

(b) Find $\vec{OB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{OB} = \dots\dots\dots (1)$$

The point P is such that APC and OPB are straight lines.(c) Using a vector method, find $\vec{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OP} = \dots\dots\dots (4)$$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 24

Vectors

1. (a) Find $\overrightarrow{AC}$

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\overrightarrow{OA} + \overrightarrow{OC} = -5\mathbf{a} + 10\mathbf{b}$$

2. Answer

$$a - 5\mathbf{a} + 10\mathbf{b}$$

3. (b) Find $\overrightarrow{OB}$

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{a} + (2\mathbf{a} + 4\mathbf{b}) = 7\mathbf{a} + 4\mathbf{b}$$

4. Answer

$$b \ 7\mathbf{a} + 4\mathbf{b}$$

1. Set up vector equations for $\overrightarrow{OP}$.

Since OPB is a straight line, $\overrightarrow{OP}$ is a scalar multiple of $\overrightarrow{OB}$. Let this be λ .

$$\overrightarrow{OP} = \lambda(7\mathbf{a} + 4\mathbf{b}) = 7\lambda\mathbf{a} + 4\lambda\mathbf{b}$$

Since P lies on the straight line AC , $\overrightarrow{AP} = \mu\overrightarrow{AC}$ for some scalar μ .

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 5\mathbf{a} + \mu(-5\mathbf{a} + 10\mathbf{b})$$

$$\overrightarrow{OP} = (5 - 5\mu)\mathbf{a} + 10\mu\mathbf{b}$$

2. Equate the coefficients of

a and b.

$$7\lambda = 5 - 5\mu \quad (1)$$

$$4\lambda = 10\mu \implies \mu = \frac{4}{10}\lambda = 0.4\lambda \quad (2)$$

3. Solve for λ .

Substitute (2) into (1):

$$7\lambda = 5 - 5(0.4\lambda)$$

$$7\lambda = 5 - 2\lambda$$

$$9\lambda = 5 \implies \lambda = \frac{5}{9}$$

5. Evaluate fraction

$$\overrightarrow{OP} = \frac{5}{9}(7\mathbf{a} + 4\mathbf{b}) = \frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

Final answer

$$\frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

EA

CLASSIFIED PROBLEM

Paper: 4WM2H

Session: Specimen

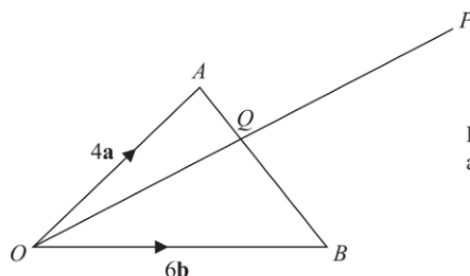
Q#: 23

Marks: 5

TOPIC: **Vectors**

Source: Specimen 4WM2H

23

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a}$$

$$\vec{OB} = 6\mathbf{b}$$

$$\vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ : QB$

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM2H | Question 23

Vectors

1. Use vectors

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

2. Use vectors

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$\overrightarrow{OP} = 4a + (2a + 8b) = 6a + 8b$$

3. Use vectors

Since O, Q, P are collinear, let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

4. Use vectors

Also Q lies on AB . Let

$$\overrightarrow{AQ} = \mu\overrightarrow{AB}$$

5. Use vectors

Then

$$\overrightarrow{AB} = 6b - 4a$$

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a)$$

$$\overrightarrow{OQ} = (4 - 4\mu)a + 6\mu b$$

6. Use vectors

Compare coefficients with

$$\lambda(6a + 8b) = 6\lambda a + 8\lambda b$$

7. Use vectors

So

$$4 - 4\mu = 6\lambda, \quad 6\mu = 8\lambda$$

8. Use vectors

From $6\mu = 8\lambda$,

$$\lambda = \frac{3}{4}\mu$$

9. Substitute

Substitute:

$$4 - 4\mu = 6 \left(\frac{3}{4}\mu \right)$$

$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

10. Use vectors

Therefore

$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

Final answer

$$AQ : QB = 8 : 9.$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jan 2020

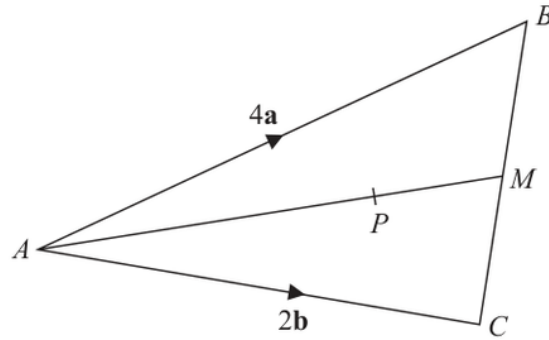
Q#: 23

Marks: 3

TOPIC: **Vectors**

Source: Jan 2020 P2H

23

Diagram **NOT**
accurately drawn

ABC is a triangle.
The midpoint of BC is M .
 P is a point on AM .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio $AP:PM$

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 23

Vectors

1. Split the ratioSince M is the midpoint of BC ,

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$\overrightarrow{AM} = \frac{4a + 2b}{2} = 2a + b$$

2. Given

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

3. Factorise

Factorise:

$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

4. Split the ratioSo AP is $\frac{3}{4}$ of AM , leaving PM as $\frac{1}{4}$ of AM .**Final answer**

$$AP : PM = 3 : 1$$

CLASSIFIED PROBLEM

Paper: P2HR

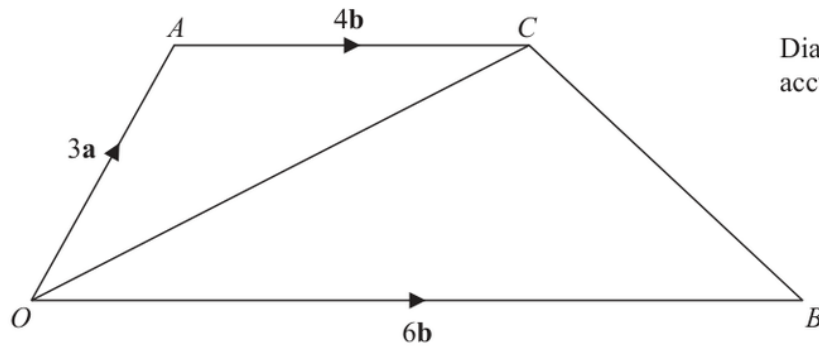
Session: Jan 2020

Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: Jan 2020 P2HR

26 The diagram shows trapezium $OACB$.Diagram **NOT**
accurately drawn

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AC} = 4\mathbf{b}$$

 N is the point on OC such that ANB is a straight line.Find $\vec{ON}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$.

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 26

Vectors

1. Read the graph

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

2. Read the graph

So

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

3. Read the graph

Point N lies on OC , so

$$\vec{ON} = t(3\mathbf{a} + 4\mathbf{b})$$

4. Read the graph

Point N also lies on AB , so

$$\vec{ON} = 3\mathbf{a} + s(6\mathbf{b} - 3\mathbf{a})$$

$$\vec{ON} = (3 - 3s)\mathbf{a} + 6s\mathbf{b}$$

5. Equate coefficients

Equate coefficients:

$$3t = 3 - 3s$$

$$4t = 6s$$

6. Read the graph

So

$$t = 1 - s$$

7. Read the graph

and

$$t = \frac{3}{2}s$$

$$\frac{3}{2}s = 1 - s$$

$$5s = 2$$

$$s = \frac{2}{5}$$

8. Read the graph

Then

$$t = \frac{3}{5}$$

9. Read the graph

Therefore

$$O\vec{N} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$O\vec{N} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

Final answer

$$O\vec{N} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

EA

CLASSIFIED PROBLEM

Paper: P1H

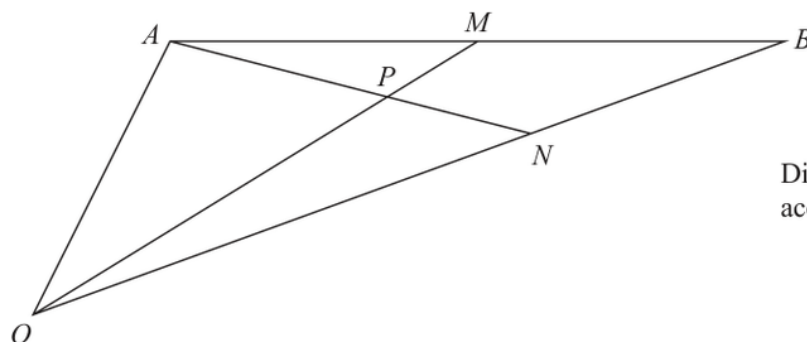
Session: Jan 2021

Q#: 23

Marks: 6

TOPIC: **Vectors**

Source: Jan 2021 P1H

23 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

 M is the midpoint of AB . N is the point on OB such that $ON:NB = 2:1$ P is the point on AN such that OPM is a straight line.Use a vector method to find $OP:PM$

Show your working clearly.

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 23

Vectors

1. Use vectors

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

2. Find the midpoint

Since M is the midpoint of AB ,

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

$$\vec{OM} = \mathbf{a} + \mathbf{b}$$

3. Use vectors

Since $ON : NB = 2 : 1$,

$$\vec{ON} = \frac{2}{3}\vec{OB}$$

$$\vec{ON} = \frac{2}{3}(2\mathbf{b}) = \frac{4}{3}\mathbf{b}$$

4. Use vectors

Point P lies on OM , so

$$\vec{OP} = t(\mathbf{a} + \mathbf{b})$$

5. Use vectors

Point P also lies on AN , so

$$\vec{OP} = 2\mathbf{a} + s\left(\frac{4}{3}\mathbf{b} - 2\mathbf{a}\right)$$

$$\vec{OP} = (2 - 2s)\mathbf{a} + \frac{4s}{3}\mathbf{b}$$

6. Use vectors

Equate coefficients of $\mathbf{a}$ and $\mathbf{b}$:

$$t = 2 - 2s$$

$$t = \frac{4s}{3}$$

7. Use vectors

So

$$\frac{4s}{3} = 2 - 2s$$

$$4s = 6 - 6s$$

$$10s = 6$$

$$s = \frac{3}{5}$$

8. Use vectors

Then

$$t = \frac{4}{5}$$

9. Use vectors

So P is $\frac{4}{5}$ of the way from O to M .

Final answer

$$OP : PM = 4 : 1$$

EA

CLASSIFIED PROBLEM

Paper: P1H

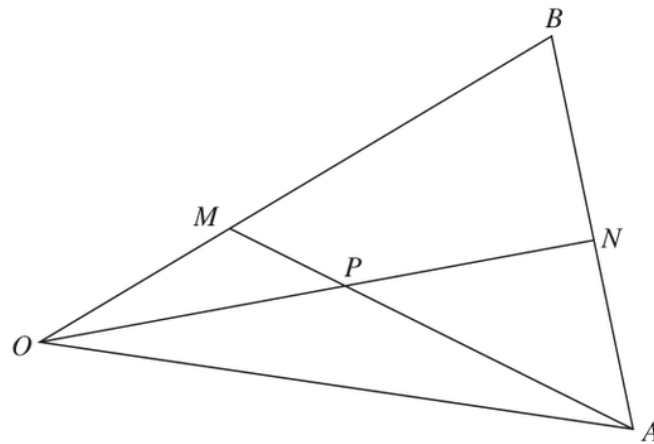
Session: Jan 2022

Q#: 22

Marks: 5

TOPIC: **Vectors**

Source: Jan 2022 P1H

22 The diagram shows triangle OAB Diagram **NOT**
accurately drawn

$$\overrightarrow{OA} = 8\mathbf{a} \quad \overrightarrow{OB} = 6\mathbf{b}$$

 M is the point on OB such that $OM:MB = 1:2$ N is the midpoint of AB P is the point of intersection of ON and AM

Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of $\mathbf{a}$ and $\mathbf{b}$
 Show your working clearly.

$$\overrightarrow{OP} = \dots\dots\dots$$

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 22

Vectors

1. Use vectors

$$\vec{OA} = 8\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}$$

2. Use vectorsSince $OM : MB = 1 : 2$,

$$\vec{OM} = \frac{1}{3}\vec{OB}$$

$$\vec{OM} = 2\mathbf{b}$$

3. Find the midpointSince N is the midpoint of AB ,

$$\vec{ON} = \frac{8\mathbf{a} + 6\mathbf{b}}{2}$$

$$\vec{ON} = 4\mathbf{a} + 3\mathbf{b}$$

4. Use vectorsPoint P lies on ON , so

$$\vec{OP} = \lambda(4\mathbf{a} + 3\mathbf{b})$$

5. Use vectorsPoint P also lies on AM , so

$$\vec{OP} = 8\mathbf{a} + \mu(2\mathbf{b} - 8\mathbf{a})$$

$$\vec{OP} = (8 - 8\mu)\mathbf{a} + 2\mu\mathbf{b}$$

6. Equate coefficients

Equate coefficients:

$$4\lambda = 8 - 8\mu$$

$$3\lambda = 2\mu$$

7. Use vectorsFrom $3\lambda = 2\mu$,

$$\mu = \frac{3}{2}\lambda$$

8. Substitute

Substitute:

$$4\lambda = 8 - 8\left(\frac{3}{2}\lambda\right)$$

$$4\lambda = 8 - 12\lambda$$

$$16\lambda = 8$$

$$\lambda = \frac{1}{2}$$

9. Use vectors

Therefore

$$\vec{OP} = \frac{1}{2}(4\mathbf{a} + 3\mathbf{b})$$

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

Final answer

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: P1HR

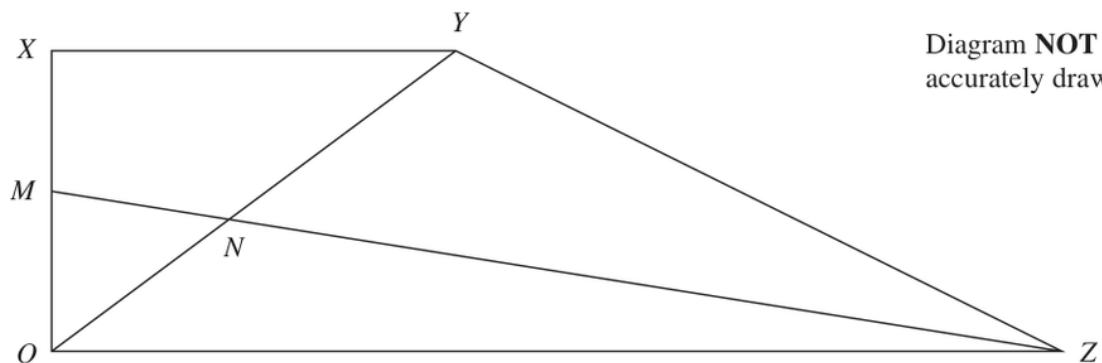
Session: Jan 2022

Q#: 24

Marks: 5

TOPIC: **Vectors**

Source: Jan 2022 P1HR

24 $OXYZ$ is a trapezium.

$$\overrightarrow{OX} = \mathbf{a}$$

$$\overrightarrow{XY} = \mathbf{b}$$

$$\overrightarrow{OZ} = 3\mathbf{b}$$

M is the midpoint of OX

N is the point such that MNZ and ONY are straight lines.

Given that $ON : OY = \lambda : 1$

use a vector method to find the value of λ

$$\lambda = \dots\dots\dots$$

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 24

Vectors

1. Take as the originTake O as the origin.

$$\vec{OX} = \mathbf{a}, \quad \vec{XY} = \mathbf{b}, \quad \vec{OZ} = 3\mathbf{b}$$

2. Use vectors

So

$$\vec{OY} = \mathbf{a} + \mathbf{b}$$

3. Find the midpointSince M is the midpoint of OX ,

$$\vec{OM} = \frac{1}{2}\mathbf{a}$$

4. Use vectorsGiven $ON : OY = \lambda : 1$,

$$\vec{ON} = \lambda(\mathbf{a} + \mathbf{b})$$

5. Use vectorsPoint N also lies on MZ , so

$$\vec{ON} = \frac{1}{2}\mathbf{a} + t\left(3\mathbf{b} - \frac{1}{2}\mathbf{a}\right)$$

$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

6. Use vectorsEquate coefficients with $\lambda\mathbf{a} + \lambda\mathbf{b}$:

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

7. Use vectors

So

$$t = \frac{\lambda}{3}$$

8. Substitute

Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2} \left(\frac{\lambda}{3} \right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

Final answer

$$\lambda = \frac{3}{7}$$

EA

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2022

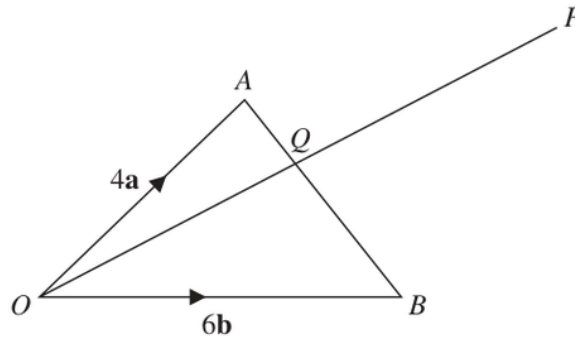
Q#: 21

Marks: 5

TOPIC: **Vectors**

Source: Jun 2022 P2H

21

Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 21

Vectors

1. Split the ratio

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

2. Split the ratio

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 4a + (2a + 8b) = 6a + 8b$$

3. Split the ratio

Since Q lies on OP , let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

4. Split the ratio

Since Q lies on AB , let

$$\overrightarrow{OQ} = \overrightarrow{OA} + \mu\overrightarrow{AB}$$

$$\overrightarrow{AB} = 6b - 4a$$

5. Split the ratio

So

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a) = (4 - 4\mu)a + 6\mu b$$

6. Equating coefficients

Equating coefficients:

$$6\lambda = 4 - 4\mu, \quad 8\lambda = 6\mu$$

7. Split the ratio

From $8\lambda = 6\mu$, $\mu = \frac{4}{3}\lambda$.

$$6\lambda = 4 - \frac{16}{3}\lambda$$

$$34\lambda = 12, \quad \lambda = \frac{6}{17}$$

$$\mu = \frac{8}{17}$$

8. Split the ratio

So $AQ : QB = \frac{8}{17} : \frac{9}{17}$.

Final answer

$$AQ : QB = 8 : 9$$

EA

CLASSIFIED PROBLEM

Paper: P2HR

Session: Jun 2022

Q#: 25

Marks: 5

TOPIC: **Vectors**

Source: Jun 2022 P2HR

25 $ABCD$ is a parallelogram and ADM is a straight line.

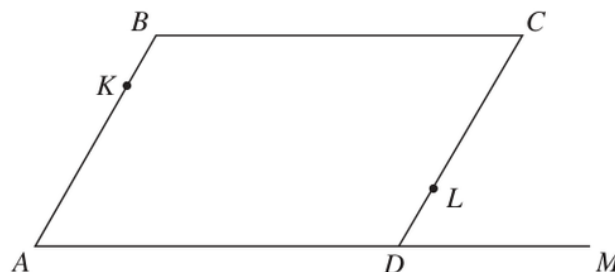


Diagram **NOT**
accurately drawn

$$\vec{AB} = \mathbf{a} \quad \vec{BC} = \mathbf{b} \quad \vec{DM} = \frac{1}{2} \mathbf{b}$$

K is the point on AB such that $AK:AB = \lambda:1$

L is the point on CD such that $CL:CD = \mu:1$

KLM is a straight line.

Given that $\lambda:\mu = 1:2$

use a vector method to find the value of λ and the value of μ

$\lambda = \dots\dots\dots$

$\mu = \dots\dots\dots$

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 25

Vectors

1. Let be the originLet A be the origin.

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

2. Read the graph

So

$$B = \mathbf{a}, \quad D = \mathbf{b}, \quad C = \mathbf{a} + \mathbf{b}$$

3. Read the graph

Also

$$D\vec{M} = \frac{1}{2}\mathbf{b}$$

4. Read the graph

So

$$M = \frac{3}{2}\mathbf{b}$$

5. Point is onPoint K is on AB :

$$K = \lambda\mathbf{a}$$

6. Read the graphPoint L is on CD , and $C\vec{D} = -\mathbf{a}$:

$$L = C + \mu(D - C)$$

$$L = \mathbf{a} + \mathbf{b} + \mu(-\mathbf{a})$$

$$L = (1 - \mu)\mathbf{a} + \mathbf{b}$$

7. Given

Given

$$\lambda : \mu = 1 : 2$$

$$\mu = 2\lambda$$

8. Read the graph

Since K, L, M are collinear, $\vec{KM}$ is a multiple of $\vec{KL}$.

$$\vec{KM} = -\lambda \mathbf{a} + \frac{3}{2} \mathbf{b}$$

$$\vec{KL} = (1 - \mu - \lambda) \mathbf{a} + \mathbf{b}$$

$$\vec{KL} = (1 - 3\lambda) \mathbf{a} + \mathbf{b}$$

9. Compare coefficients using

Compare coefficients using

$$\vec{KM} = \frac{3}{2} \vec{KL}$$

$$-\lambda = \frac{3}{2}(1 - 3\lambda)$$

$$-2\lambda = 3 - 9\lambda$$

$$7\lambda = 3$$

$$\lambda = \frac{3}{7}$$

10. Read the graph

Then

$$\mu = 2\lambda = \frac{6}{7}$$

Final answer

$$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Jun 2024

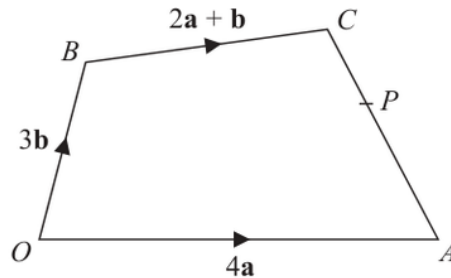
Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Jun 2024 P2H

24

Diagram **NOT**
accurately drawnThe diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

The point P lies on AC such that $AP:PC = 3:2$ The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 24

Vectors

1. Read the graph

$$\vec{OA} = 4\mathbf{a}, \quad \vec{OB} = 3\mathbf{b}, \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

2. Read the graph

So

$$\vec{OC} = \vec{OB} + \vec{BC}$$

$$\vec{OC} = 2\mathbf{a} + 4\mathbf{b}$$

3. Read the graph

For part (a),

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$\vec{AC} = (2\mathbf{a} + 4\mathbf{b}) - 4\mathbf{a}$$

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$$

4. Read the graph

For part (b), $AP : PC = 3 : 2$, so

$$\vec{OP} = \vec{OA} + \frac{3}{5}\vec{AC}$$

$$\vec{OP} = 4\mathbf{a} + \frac{3}{5}(4\mathbf{b} - 2\mathbf{a})$$

$$\vec{OP} = \frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

5. Read the graph

Since O, P, Q are collinear,

$$\vec{OQ} = t\vec{OP}$$

$$\vec{OQ} = t \left(\frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b} \right)$$

6. Read the graph

Since B, C, Q are collinear,

$$\vec{OQ} = \vec{OB} + s\vec{BC}$$

$$\vec{OQ} = 3\mathbf{b} + s(2\mathbf{a} + \mathbf{b})$$

$$\vec{OQ} = 2s\mathbf{a} + (3 + s)\mathbf{b}$$

7. Equate coefficients

Equate coefficients:

$$\frac{14}{5}t = 2s$$

$$\frac{12}{5}t = 3 + s$$

8. Solve equation

From the first equation,

$$s = \frac{7}{5}t$$

9. Substitute into the second

Substitute into the second:

$$\frac{12}{5}t = 3 + \frac{7}{5}t$$

$$t = 3$$

10. Read the graph

Therefore

$$\vec{OQ} = 3\vec{OP}$$

$$\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

Final answer

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}, \vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: P2HR

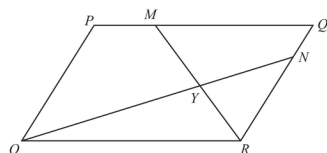
Session: Jun 2024

Q#: 25

Marks: 6

TOPIC: **Vectors**

Source: Jun 2024 P2HR

25 $OPQR$ is a parallelogram.Diagram NOT
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answers in simplest form

(i) $\vec{ON}$

(1)

(ii) $\vec{MR}$

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ _____
(4)

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 25

Vectors

1. Calculate value

$$\overrightarrow{OP} = 2a, \quad \overrightarrow{OR} = 3b$$

2. Calculate value

So

$$\overrightarrow{OQ} = 2a + 3b$$

3. Evaluate fraction

Since $PM : PQ = 1 : 4$,

$$\overrightarrow{PM} = \frac{1}{4} \overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

4. Evaluate fraction

Since $RN : RQ = 4 : 5$,

$$\overrightarrow{RN} = \frac{4}{5} \overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

5. Evaluate fraction

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

6. Let be on

Let Y be on ON , so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

7. Evaluate fraction

Also Y is on MR , so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

8. Evaluate fraction

Compare coefficients of a and b :

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

9. Solve equation

From the first equation,

$$t = 1 - \frac{4}{5}k$$

10. Substitute into the second equation

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4}\left(1 - \frac{4}{5}k\right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

Final answer

$$\overrightarrow{ON} = \frac{8}{5}a + 3b, \overrightarrow{MR} = -2a + \frac{9}{4}b, \text{ and } k = \frac{5}{8}$$

CLASSIFIED PROBLEM

Paper: P1H

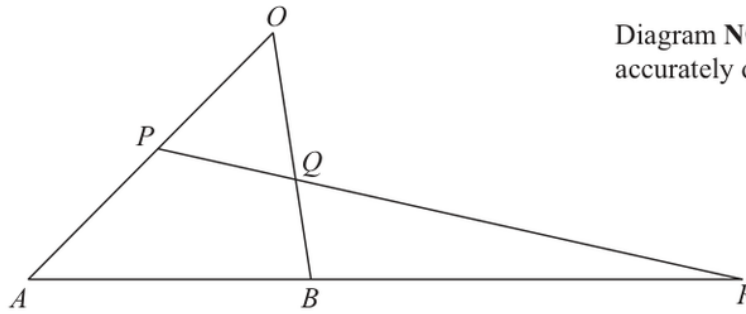
Session: Jun 2025

Q#: 23

Marks: 5

Topic: **Vectors**

Source: Jun 2025 P1H

23Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(1)

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots \quad (4)$$

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 23

Vectors

1. Use vectors

For part (a),

$$\vec{OA} = 12\mathbf{a}, \quad \vec{OB} = 8\mathbf{b}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}$$

2. Find the midpointFor part (b), P is the midpoint of OA , so

$$\vec{OP} = 6\mathbf{a}$$

3. Given

Given

$$AB : BR = 1 : 2$$

$$\vec{BR} = 2\vec{AB}$$

$$\vec{BR} = 16\mathbf{b} - 24\mathbf{a}$$

4. Use vectors

Therefore

$$\vec{OR} = \vec{OB} + \vec{BR}$$

$$\vec{OR} = 8\mathbf{b} + 16\mathbf{b} - 24\mathbf{a}$$

$$\vec{OR} = -24\mathbf{a} + 24\mathbf{b}$$

5. Use vectorsSince P, Q, R are collinear, write Q on PR :

$$\vec{OQ} = \vec{OP} + t(\vec{OR} - \vec{OP})$$

$$\vec{OQ} = 6\mathbf{a} + t(-30\mathbf{a} + 24\mathbf{b})$$

$$\vec{OQ} = (6 - 30t)\mathbf{a} + 24t\mathbf{b}$$

6. Use vectors

But Q lies on OB , so there is no $\mathbf{a}$ component:

$$6 - 30t = 0$$

$$t = \frac{1}{5}$$

7. Use vectors

Then

$$\vec{OQ} = 24 \left(\frac{1}{5} \right) \mathbf{b}$$

$$\vec{OQ} = \frac{24}{5} \mathbf{b}$$

8. Use vectors

Since $\vec{OQ} = n\mathbf{b}$,

$$n = \frac{24}{5}$$

Final answer

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}, \quad n = \frac{24}{5}$$

CLASSIFIED PROBLEM

Paper: P1H

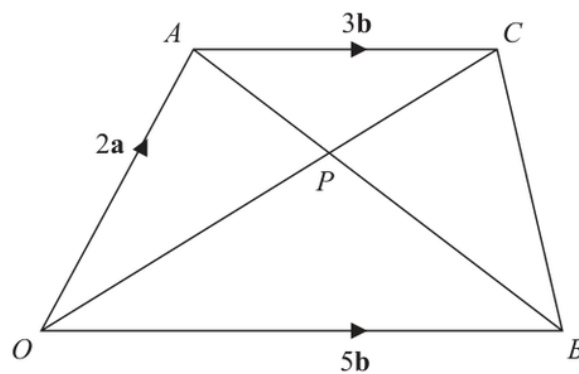
Session: May 2021

Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: May 2021 P1H

26 $OACB$ is a trapezium.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals, OC and AB , of the trapezium intersect at the point P .Find and simplify an expression, in terms of $\mathbf{a}$ and $\mathbf{b}$, for $\vec{OP}$
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 26
Vectors

1. Use vectors

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

2. Use vectors

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

3. Use vectors

Since P lies on OC , let

$$\overrightarrow{OP} = t(2\mathbf{a} + 3\mathbf{b})$$

4. Use vectors

Since P also lies on AB , let

$$\overrightarrow{OP} = \overrightarrow{OA} + s\overrightarrow{AB}$$

5. Use vectors

Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

6. Use vectors

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

7. Use vectors

Compare this with

$$t(2\mathbf{a} + 3\mathbf{b}) = 2t\mathbf{a} + 3t\mathbf{b}$$

8. Equating coefficients

Equating coefficients:

$$2t = 2 - 2s$$

$$3t = 5s$$

9. Use vectors

From $t = 1 - s$,

$$3(1 - s) = 5s$$

$$3 = 8s$$

$$s = \frac{3}{8}$$

10. Use vectors

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

11. Use vectors

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

Final answer

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: P2H

Session: Nov 2023

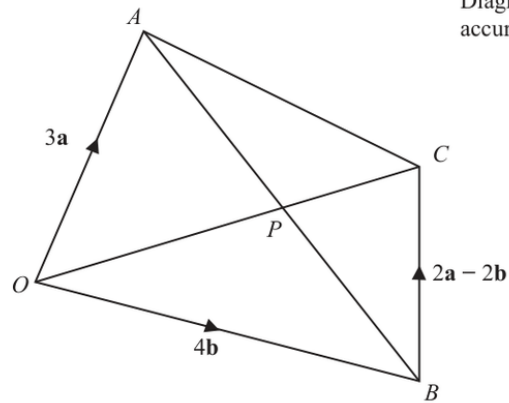
Q#: 22

Marks: 5

TOPIC: **Vectors**

Source: Nov 2023 P2H

22

Diagram **NOT**
accurately drawn $OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots (1)$$

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots (1)$$

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

.....
(3)

.....
(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 22

Vectors

1. Split the ratio

$$\overrightarrow{OA} = 3a, \quad \overrightarrow{OB} = 4b, \quad \overrightarrow{BC} = 2a - 2b$$

2. Part (a)

Part (a):

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a$$

3. Split the ratio

For part (b), let P divide AB so that

$$\overrightarrow{AP} = \mu \overrightarrow{AB}$$

4. Split the ratio

Then

$$\overrightarrow{OP} = \overrightarrow{OA} + \mu \overrightarrow{AB} = 3a + \mu(4b - 3a) = (3 - 3\mu)a + 4\mu b$$

5. Split the ratio

Since P lies on OC , also let

$$\overrightarrow{OP} = \lambda \overrightarrow{OC} = \lambda(2a + 2b)$$

6. Equating coefficients

Equating coefficients:

$$3 - 3\mu = 2\lambda, \quad 4\mu = 2\lambda$$

7. Split the ratio

From $4\mu = 2\lambda$, $\lambda = 2\mu$.

$$3 - 3\mu = 4\mu$$

$$\mu = \frac{3}{7}$$

8. Split the ratio

So $AP : PB = \frac{3}{7} : \frac{4}{7}$.

Final answer $\vec{OC} = 2a + 2b$, $\vec{AB} = 4b - 3a$, and $AP : PB = 3 : 4$

EA

CLASSIFIED PROBLEM

Paper: P1H

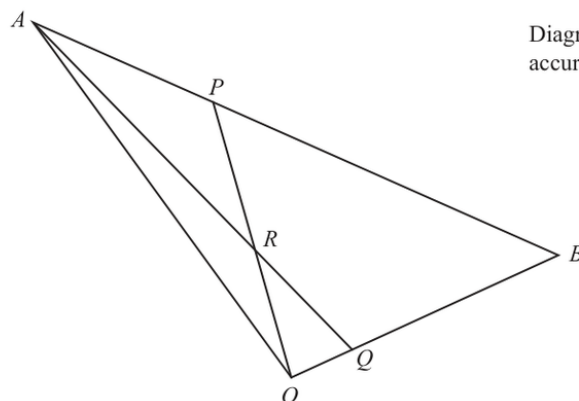
Session: Nov 2024

Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Nov 2024 P1H

24 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

(iii) $\vec{OR}$

.....
(4)

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 24
Vectors

1. Read the graph

$$\vec{OA} = 10\mathbf{a}, \quad \vec{OB} = 10\mathbf{b}$$

2. Read the graph

Also,

$$\vec{AP} = \frac{1}{4}\vec{AB}$$

3. Read the graph

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

4. Read the graph

For part (i),

$$\vec{OQ} = 2\mathbf{b}$$

$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

5. Read the graph

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{OP} = 10\mathbf{a} - \frac{5}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

$$\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

6. Read the graph

For part (iii), R lies on OP , so

$$\vec{OR} = t\vec{OP}$$

7. Read the graph

It also lies on AQ , so

$$\vec{OR} = \vec{OA} + s\vec{AQ}$$

$$\vec{OR} = 10\mathbf{a} + s(2\mathbf{b} - 10\mathbf{a})$$

$$\vec{OR} = (10 - 10s)\mathbf{a} + 2s\mathbf{b}$$

8. Read the graph

Equate with

$$t\left(\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}\right)$$

$$\frac{15}{2}t = 10 - 10s$$

$$\frac{5}{2}t = 2s$$

9. Solve equation

From the second equation,

$$s = \frac{5}{4}t$$

10. Substitute

Substitute:

$$\frac{15}{2}t = 10 - \frac{25}{2}t$$

$$20t = 10$$

$$t = \frac{1}{2}$$

11. Read the graph

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

Final answer

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}, \vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}, \vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2025

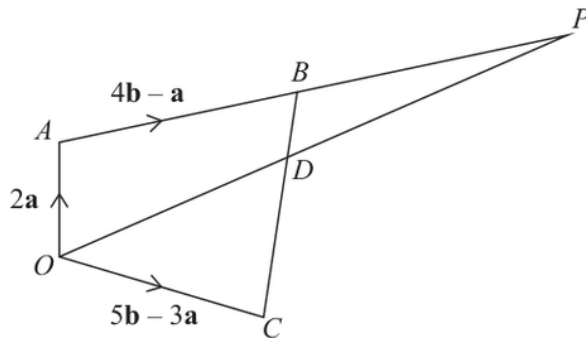
Q#: 27

Marks: 6

TOPIC: **Vectors**

Source: Nov 2025 P1H

27

Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$
 Simplify your answer.

(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

$$n = \dots\dots\dots$$

(4)

(Total for Question 27 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 27

Vectors

1. Read the graph

$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

2. Read the graph

First find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

3. Read the graph

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

4. Read the graph

For part (b), $BD : DC = 1 : 3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

5. Read the graph

Since A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

6. Read the graph

But O, D, P are collinear, and $\vec{OD}$ has only a $\mathbf{b}$ component.

7. Read the graph

So

$$2 - s = 0$$

$$s = 2$$

8. Read the graph

Then

$$\vec{OP} = 8\mathbf{b}$$

9. Given

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

Final answer

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$